

MANY TREATMENTS, SOME INSTRUMENTS: SHARP IDENTIFICATION AND SPECIFICATION TESTING IN TREATMENT SELECTION MODELS*

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PRELIMINARY DRAFT

I present new results on partial identification and specification testing in nonparametric instrumental variable (IV) models with (potentially continuous and/or unbounded) outcomes, discrete (ordered or unordered) treatments, and discrete randomly assigned instruments. Other than random assignment of the instruments, I do not require any particular restrictions on how instruments influence selection into treatment; instead, I develop a unified (partial) identification approach that allows researchers to flexibly choose the target parameter and treatment selection (or choice) model of interest. The proposed approach sharply incorporates the full identifying power of randomly assigned instruments, and can tractably accommodate models with potentially infinite dimensional unobserved heterogeneity without relying on any parametric or semi-parametric assumptions. Using a novel geometric characterization that is less “black box” than purely computational approaches, I characterize the empirical content and testable implications of such models, and obtain tractable representations of the identified set for counterfactual choices, marginal and joint outcome distributions, outcome means, and treatment effects. For a broad class of such models, I suggest methods for performing inference on mean causal effects and counterfactual outcomes, and for specification testing treatment selection models. Finally, for cases where analytical approaches are infeasible, I propose a computationally feasible algorithm for implementing the proposed methods.

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1. Introduction

I study nonparametric instrumental variable (IV) models with multivalued treatments, randomly assigned discrete instruments, and heterogeneous agents. Causal relationships in such models can (potentially after conditioning on baseline covariates) be represented by the following model:

$$\text{Treatment Selection Equation: } D = g_D(Z, \mathbf{V}), \quad (1)$$

$$\text{Outcome Equation: } Y = g_Y(D, \mathbf{V}, \varepsilon_Y), \quad (2)$$

where g_D and g_Y are (generally unknown) functions, Y is the observed (potentially continuous) outcome of interest, D is the observed (discrete-valued) treatment status, Z is the observed (discrete-valued) assigned instrument value, and $\mathbf{V}$ and ε_Y are (potentially infinite dimensional) unobservables representing unobserved heterogeneity; $Z, \mathbf{V}$ and ε_Y are mutually independent exogenous variables, which implies random assignment of the instrument Z . I require that D and Z are finitely-supported, but the support of Y is only required to be a compact convex subset of $\mathbb{R}^n$ (or finite). I show that, under these basic requirements, this model is formally equivalent to the usual potential outcomes IV model. The framework in this paper does not *require* any further restrictions on the general model above, but allows the analyst to flexibly (and sharply) incorporate additional structure: I focus on models that impose assumptions on the treatment selection equation (1) but remain agnostic about the outcome equation (2).¹ For such models, this paper provides characterizations of

- (i) their empirical content and (sharp) testable implications,
- (ii) the identified set for a large class of target parameters, including mean counterfactuals² such as counterfactual choice probabilities, mean outcomes, and mean treatment effects,
- (iii) the identified set for the joint distribution of potential outcomes.

Since D and Z are only required to be finitely-supported, the results in this paper apply to settings with multiple dimensions of treatment (e.g., mediation analysis) and/or unobserved heterogeneity. Indeed, the framework of this paper applies to settings with “multiple” instruments or treatments just as easily as it does to settings with “multivalued” instruments or treatments: such distinctions are incorporated via restrictions on the selection equation. Moreover, by allowing Z (or some components of it) to be included in D , the results also apply to models where conventional IV exclusion restrictions may be violated.

Based on these results, I develop a fundamentally nonparametric approach to the partial identification of a large class of causal parameters in such models. The proposed approach sharply incorporates the full identifying power of randomly assigned instruments, and can tractably accommodate models with even infinite dimensional unobserved heterogeneity without relying on any parametric or semi-parametric assumptions. The unified approach of this paper allows researchers to flexibly choose the target parameter and selection model most relevant for their policy question and setting, without requiring new derivations, even when such a model/parameter has not been previously considered in the methodological literature. The proposed approach can also be used to derive analytic expressions for bounds on causal parameters, which helps provide intuition for the source and structure of identifying power in such models.

Identification in this setting is hindered by the presence of the unobservable confounder $\mathbf{V}$, which can generate selection bias. As in Heckman and Pinto (2018), this paper’s approach to

¹Although the focus is on selection models, the proposed framework can sharply incorporate (and test) certain assumptions on (potentially conditional) mean outcomes. For example, some assumptions of the form $E(Y_d) \geq E(Y_{d'})$ can be incorporated, but generally not assumptions of the form $P(Y_d \geq Y_{d'}) = 1$. See Remark 3.9.

²Where the mean can be taken over the whole population, or over particular subpopulations of interest.

the selection bias problem uses the instrument Z to partition the support of V into cells that induce the same vector of counterfactual treatment assignments, or *response-type*.³ I show that assumptions on g_D and/or V have identifying power *only* through restrictions on the relative frequencies of different response-types (including, in particular, which response-types are eliminated by the model). Heckman and Pinto (2018) showed that the problem of identifying response-type conditional counterfactuals can be represented as a finite mixture model where the mixture components are the conditional distributions generated by response-types. I attack this mixture problem directly, leveraging a geometric characterization of such mixture models to develop new results on the partial identification of components of overlapping mixtures, which may be of independent interest for a variety of other problems.

The identification results in this paper are sharp in the sense that they exhaust all of the information in the model and in the available data. Importantly, these results are sharp under the empirically and theoretically common assumption of random assignment (“full” independence) of the instrument, as opposed to the weaker assumption of mean independence considered in some of the related literature on partial identification in IV models (as in, e.g., Manski (1990); Mogstad, Santos, and Torgovitsky (2018)). This distinction can be important, since the stronger full independence assumption can lead to tighter bounds on target parameters, as discussed below. Moreover, as noted by Mogstad and Torgovitsky (2018), it is difficult to think of practical applications where mean independence holds but full independence fails to hold.

The main focus of this paper is identification, but I also consider approaches for estimation and inference based on the above identification results. In particular, for a broad class of such models, I outline specification tests for treatment selection models, and inclusion tests for mean counterfactuals and treatment effects that can be inverted to construct confidence regions. These tests can be implemented using, e.g., Chernozhukov, Lee, and Rosen (2013). For cases where analytical approaches are infeasible, I propose a computationally feasible nonparametric algorithm for constructing estimators for the proposed approach, based on computing the support function of suitable convex sets. The proposed algorithm will always find an exact solution finitely many steps, but may be intractable when the number of instruments and/or response-types is large; in such cases, the proposed algorithm can be used to tractably approximate the quantities of interest using a generalized primal-dual approach that produces deterministic, two-sided, bounds on the approximation error. The resulting “outer” approximation can be used for conservative inference, whereas the “inner” approximation can be used to perform “optimistic” inference which, while invalid, is useful for determining whether there are any inferential returns to improving the approximation.

1.1. Related Literature. This paper contributes to a very large literature on identification in heterogeneous agent IV models, I provide an overview of the most relevant papers here. Much of the literature on IV models with multiple treatments and/or instruments have focused on point identification under relatively restrictive assumptions on treatment selection: Angrist and Imbens (1995) assume “monotonicity” of the instrument and extend Imbens and Angrist (1994)’s results to a setting with multiple levels of (ordered) treatments, Heckman, Urzua, and Vytlacil (2008) considers multiple treatments within the Marginal Treatment Effect (MTE) framework of Heckman and Vytlacil (1999, 2001, 2007a,b) but rely on continuous monotonic instruments, Heckman and Pinto (2018) consider point identification in the same setting as this paper and propose the “Unordered Monotonicity” condition which

³Sometimes referred to as “principal strata”, “complier-type”, or “response-group” in the related literature, the notion of response-types used in this paper is the same as Balke and Pearl (1994), Imbens and Angrist (1994), Frangakis and Rubin (2002), Gu and Russell (2022), Lee and Salanié (2023), and many other papers.

allows several mean counterfactuals and treatment effects to be point-identified, Mountjoy (2022) provides point identification for some parameters of interest using continuous instruments and the less restrictive “Unordered Partial Monotonicity” assumption, which builds on the “Partial Monotonicity” assumption introduced by Mogstad, Torgovitsky, and Walters (2021). Lee and Salanié (2018) consider a broad class of treatment selection models based on combinations of threshold crossing rules and develop point identification results that rely on continuous instruments. When only finitely supported instruments are available, the partial identification framework of this paper can sharply incorporate any of these assumptions on treatment selection.⁴

More closely related to this paper is the growing literature on partial identification in IV models, as considered by Manski (1989), Manski and Pepper (2000, 2009), Shaikh and Vytlacil (2011), Mourifié (2015), Torgovitsky (2019b), among others. The literature on sharp partial identification with randomly assigned (“fully independent”) instruments has focused primarily on the case where the treatment is binary and the instrument satisfies monotonicity. For settings where the outcome, treatment, and instrument are all binary, Balke and Pearl (1997) provides sharp bounds for the Average Treatment Effect (ATE), with and without imposing monotonicity of the instrument. For general discrete or continuous outcomes, in the case where the treatment and instrument are both binary, Kitagawa (2021) derives the sharp identification region for potential outcome distributions and provides analytic expressions for sharp bounds on the ATE, where the instrument may or may not be monotonic. Similar in spirit to the current paper, in the binary treatment and monotonic instrument setting, Marx (2024) goes beyond the ATE and provides sharp analytic bounds for a large class of policy relevant treatment effects. Han and Yang (2024) proposes a computational approach for similar parameters in the same setting as Marx (2024), while also allowing for discrete instruments that do not satisfy the exclusion restriction.

Almost all of the related sharp partial identification results in IV settings with multiple treatments assume that the instrument satisfies an exogeneity condition (random assignment being the strongest such assumption) but do not consider further restrictions on selection into treatment. For example, Manski (2003) assumes that the instrument and outcomes satisfy a marginal statistical independence condition, and provides a sharp characterization of potential outcome distributions for settings with many treatments and a finitely supported instrument. Similarly, Beresteanu, Molchanov, and Molinari (2012) assume random assignment of the instrument, and use random set theory to develop sharp identification results for settings with multiple treatments and multivalued (potentially continuous) instruments, extending the results of Manski (2003), Balke and Pearl (1997), and Kitagawa (2021); the notion of response-types is not present in their analysis, however, and they do not consider monotonicity or any other assumptions on treatment selection.⁵ An exception is Cheng and Small (2006), who consider a setting with binary outcomes, three treatments and three instrument values, and provide sharp bounds on response-type conditional treatment effects for particular forms of one-sided noncompliance.

⁴In order to directly apply the identification results in this paper, the researcher needs to derive the *a priori* restrictions on the relative frequencies of response-types imposed by the selection model of interest. In the many leading cases where the selection model simply *eliminates* some response-types (e.g., Ordered/Unordered/Vector/Partial Monotonicity, and the models considered in Lee and Salanié (2018)), this information is generally simple to derive. More generally, although this information is always “known” from the model, it may be non-trivial to derive for complicated selection models.

⁵Interestingly, although this paper’s approach to identification does not involve random sets, the *Minkowski integral* operation from convex geometry (a generalization of the Minkowski sum), used extensively in this paper, is related to the notion of *Aumann expectation* from random set theory, used in Beresteanu et al. (2012) and many other papers. See Remark 3.7 for a brief discussion of the connection between the two approaches.

Several recent papers consider computational approaches for sharp partial identification under the weaker assumption of mean independence of the instrument. In the case where both the treatment and instrument are binary, and under mean independence of the instrument, Manski (1990) provides bounds on the ATE, and Huber, Laffers, and Mellace (2017) provides bounds on response-type conditional treatment effects, with and without monotonicity of the instrument. Also in the binary treatment setting, but allowing the mean independent monotonic instrument to be non-binary, Mogstad et al. (2018) (hereafter, MST) propose a computational approach for partial identification of the large class of target parameters that can be expressed as weighted averages of the MTE function, and their approach was recently extended to selection models with multiple mean independent monotonic instruments (i.e. partial monotonicity) by Mogstad, Torgovitsky, and Walters (2023). Similarly, Gu and Russell (2023) propose a computational approach that can be used for counterfactual inference in certain multiple treatment IV models under mean independence. While these approaches can find the set of values for the target parameter that are consistent with point-identified mean outcomes, and thereby exhaust the identifying power of the mean independence assumption, they do not ensure that the values of the target parameter are consistent with the conditional *distributions* of outcomes that are point-identified when the instruments are fully independent (i.e. randomly assigned). In their respective settings, these approaches complement the results in this paper since they may be applicable even when full independence of the instrument is not a credible assumption and, in the case of the last three papers, can incorporate instruments with infinite/continuous support and stronger assumptions on the outcome equation. When the instrument is randomly assigned, however, incorporating the additional consistency requirements implied by full independence of the instrument can lead to strictly tighter nonparametric bounds on target parameters, even in the basic binary treatment case (for examples, see discussions in Balke and Pearl (1997), Marx (2024), and Han and Yang (2024)).

Two contemporaneous papers, Lee and Salanié (2023) and Kamat, Norris, and Pecenco (2023), consider partial identification in multiple treatment IV models with randomly assigned discrete-valued instruments. Unlike this paper, however, both papers propose approaches that require substantial restrictions on unobserved heterogeneity. Moreover, although they assume that the instrument is randomly assigned, they propose approaches that do not utilize the additional identifying power provided by full independence of the instrument. Lee and Salanié (2023) consider models where treatment selection is governed by an additive random-utility model and the discrete-valued instruments satisfy a “targeting” condition, focusing on two special cases: the binary instrument case, and the case with three instrument levels and three treatments. For these cases, they derive bounds on response-type probabilities and, under additional assumptions, bounds on certain mean outcomes and treatment effects. Their bounds are not sharp in general, but have the attractive property of being analytic and simple to compute. Kamat et al. (2023) propose a computational approach for bounding a broad class of target parameters in the class of models considered by Lee and Salanié (2018), based on extending the arguments in MST to settings with multiple dimensions of unobserved heterogeneity. They rely on the additional assumption that the dependence between different dimensions of unobserved heterogeneity can be parameterized by a finite dimensional parameter, and propose a procedure that applies MST’s approach at every point in the parameter space. Their bounds are generally not sharp under random assignment of the instrument.⁶

⁶Although they assume full independence of the instrument, their approach is based on generalizing the arguments in MST (which only rely on mean independence), and does not exploit the additional information available under their stronger assumption of full independence.

This paper is also related to the literature on specification testing in IV models. This paper extends the results of Kitagawa (2015) and Mourifié and Wan (2017) on sharp testable implications of LATE assumptions to treatment selection models beyond monotonicity and to the general multivalued treatment and instrument setting. As special cases, this paper also provides a sharp test for random assignment, for which a test was proposed by Kédagni and Mourifié (2020), and also for Unordered Monotonicity, for which a test was proposed by Sun (2023). Neither of these papers show sharpness under full instrument independence, however, unlike this paper.⁷ The framework of this paper can also be used to perform sensitivity analysis in the spirit of Noack (2021), but I do not pursue this formally in this draft. This paper is also related to the literature on the identification of joint distribution of potential outcomes, as investigated by Heckman and Smith (1995, 1998), Heckman, Smith, and Clements (1997), and more recently by Mourifié, Henry, and Méango (2020), Jun and Lee (2023) and Kaji and Cao (2023); the approach of this paper is fully nonparametric and in a relatively more general setting. Finally, this paper is related to the literature on nonparametric identification of mixture components, as in Horowitz and Manski (1995, 1998, 2000) and Henry, Kitamura, and Salanié (2014), and to the literature on nonparametric sample selection, as in Lee (2009). For several special cases, the analytic results derived in this paper share features with the analytic results of Horowitz and Manski (1995), Lee (2009), Kitagawa (2021), Kaji and Cao (2023), and Marx (2024).

Organization of the paper. The remainder of this paper is organized as follows. Section 3 formalizes the analytical framework and setting, including the notion of “treatment selection model”, and develops some high level results about the structure of identifying power in treatment selection models. Section 3.2 introduces the geometric characterization of IV mixture models and uses it to derive the identified set for response-type probabilities and obtains, as a by-product, a sharp specification test for treatment selection models, and characterizations of the identified set for marginal and joint outcome distributions, outcome means, and treatment effects. Section 4 outlines an inference strategy for implementing the proposed specification tests and develops an inclusion test for counterfactual means and treatment effects. Section 5 discusses potential computational challenges that may complicate practical implementation, and presents a generally applicable computational algorithm that can be used to implement the proposed methodology. Section 6 concludes. In Appendices A to D I provide proofs of results not proved in the main text, and in Appendix E I formally state and prove the equivalence of the structural IV model of this paper and the usual potential outcomes IV model.

Notation. For any set K , $|K|$ denotes its cardinality. For any number $n \in \mathbb{N}$, $[n] := \{1, \dots, n\}$. For any event A , denote by $\mathbb{1}\{A\}$ the indicator function takes value 1 if A is true, and 0 otherwise. For any random variable X , $\text{Law}(X)$ denotes its distribution, and $\text{supp}(X)$ denotes its support. For any distribution F of some multi-dimensional variable $X = (X_1, \dots, X_K)$: (i) $\text{supp}(F) := \text{supp}_F(X)$ denotes the support of X under $X \sim F$, (ii) F_{X_j} denotes the marginal distribution of X_j under $X \sim F$, (iii) $F_{X_j|X_k=x_k}$ denotes the conditional distribution of X_j , given $X_k = x_k$, under $X \sim F$.

2. Setup and Motivating Examples

2.1. Basic Framework and Setting. We will work with a complete non-atomic probability space, (Ω, Σ, P) , on which all random variables in this paper are defined. All of the analysis

⁷The results in this paper should allow us to resolve the conjectured sharpness of these papers, but I have not yet pursued this.

in this section and the next will treat the joint distribution of the observed data, (Y, D, Z) , as known. We may also observe a vector of baseline covariates that are unaffected by treatment, and the analysis in this paper should be understood as holding conditional on such variables. Throughout, we will assume that the observed data satisfies the following regularity conditions.

Condition D (Regularity Conditions). *The support of D , denoted $\mathcal{D}$, and the support of Z , denoted $\mathcal{Z}$, are finite. The support of Y , denoted $\mathcal{Y}$, is either a finite set or a compact and convex subset of $\mathbb{R}^n$. Moreover, $\mathcal{Y}$ is equipped with the Borel σ -algebra $\mathcal{B}_Y$ and a probability measure μ . For all $d \in \mathcal{D}$ and $z \in \mathcal{Z}$, $E[Y \mid D = d, Z = z]$ exists and $|E[Y \mid D = d, Z = z]| < \infty$. Finally, for all $z \in \mathcal{Z}$, $P(Z = z) > 0$.*

All inequalities and equalities involving density functions should be interpreted as holding μ -almost-everywhere. When I say “all” or “each” $y \in \mathcal{Y}$, I mean “ μ -almost-all”, and when I say that a property holds for “some $y \in \mathcal{Y}$ ”, or that “there exists $y \in \mathcal{Y}$ ” with some property, I mean that the set of $y \in \mathcal{Y}$ with the property has positive μ measure. Similarly, all identification results involving density functions should be understood to hold only up to a zero μ -measure subset of $\mathcal{Y}$.

The basic setting is summarized in the following condition:⁸

Condition M (Structural IV Model). *There exists random variables V and ε_Y defined on (Ω, Σ, P) that satisfy the two following properties:*

- (i) *There exist measurable functions $g_D : \mathcal{Z} \times \text{supp}(V) \rightarrow \mathcal{D}$ and $g_Y : \mathcal{D} \times \text{supp}(V) \times \text{supp}(\varepsilon_Y) \rightarrow \mathcal{Y}$ such that, almost surely,*

$$D = g_D(Z, V), \quad (1)$$

$$Y = g_Y(D, V, \varepsilon_Y). \quad (2)$$

and, for each $d \in \mathcal{D}$, $g_Y(d, V, \varepsilon_Y)$ is μ -integrable and has a density function with respect to μ .

- (ii) *The random variables Z, V, ε_Y are mutually independent.*

Condition M introduces the unobservables (V, ε_Y) and describes the causal relationship among the variables $(Y, D, Z, V, \varepsilon_Y)$. Condition M(i) makes explicit the structural equations (1)-(2) that are used below for formally defining and analyzing treatment selection models. Condition M(i) also imposes an integrability condition on g_Y to ensure that the relevant conditional expectations of Y exist.⁹ The exogeneity requirement on Z is captured by Condition M(ii), which implies that Z is randomly assigned. Condition M(ii) also imposes that the (potentially infinite-dimensional) unobservables V and ε_Y are independent from each other, which is an important feature for many of the results presented below. Informally, this assumption ensures that the residual uncertainty (i.e., after conditioning on V) in the outcome is (allowed to be) rich enough so that assumptions on g_D and V alone cannot restrict the outcome equation “too much”. In particular, Condition M(ii) limits the role played by V in restricting the *joint* distribution of potential outcomes.

⁸This is the model considered in Heckman and Pinto (2018), but the formulation here is slightly different: in Condition M, g_D and g_Y are deterministic functions, whereas Heckman and Pinto (2018) model treatment selection as driven by a random function of Z and V , and require the introduction of an independent shock ε_D to represent D as a deterministic function, $\tilde{g}_D(Z, V, \varepsilon_D)$; such models still satisfy Condition M, however, since we can just take $g_D(Z, V) := \int_{\text{supp}(\varepsilon_D)} \tilde{g}_D(Z, V, e) d\mu(e)$. I use this formulation because it is convenient for simplifying the formal definition of treatment selection models, introduced in the next subsection.

⁹Condition M(i) also imposes that $g_Y(d, V, \varepsilon_Y)$ has support $\mathcal{Y}$, for all $d \in \mathcal{D}$, but all results easily extend to settings where $\mathcal{Y}$ depends on d .

The structural IV model described above is equivalent to the usual potential outcomes model with random assignment of the instrument.¹⁰ Indeed, potential outcomes can be defined, for $d \in \mathcal{D}$ and $z \in \mathcal{Z}$, as the random variables

$$\begin{aligned} D_z &:= g_D(z, \mathbf{V}) , \\ Y_d &:= g_Y(d, \mathbf{V}, \varepsilon_Y) . \end{aligned}$$

In this notation,

$$\begin{aligned} D &= \sum_{z \in \mathcal{Z}} \mathbb{1}\{Z = z\} D_z , \\ Y &= \sum_{d \in \mathcal{D}} \mathbb{1}\{D = d\} Y_d , \end{aligned}$$

and we have that Condition M(ii) implies $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$, which is the usual random assignment assumption. I will use both notations, as convenient for the exposition. I now introduce some notions that will be useful for formally studying identification in this setting.

Parent distributions. In the remainder, distributions of $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z}), Z)$ will be called “parent” distributions and distributions of (Y, D, Z) will be called “observed data” distributions.

Denote by $\mathcal{M}$ the class of parent distributions that are consistent with Condition D and Condition M; i.e. all joint distributions of these variables that (i) have the correct supports and integrability properties and (ii) satisfy $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$. In other words, a distribution Q is in $\mathcal{M}$ if and only if it witnesses Condition M for *some* distribution of (Y, D, Z) that satisfies the regularity conditions given by Condition D.

For any $Q \in \mathcal{M}$, denote by P_Q the distribution of the observed data (Y, D, Z) under the parent distribution Q . Throughout the remainder, P will denote the (known) observed data distribution. Let $\mathcal{Q}(P) \subseteq \mathcal{M}$ be the identified set for parent distributions under Condition M; formally, $\mathcal{Q}(P) := \{Q \in \mathcal{M} \mid P_Q = P\}$. We will generally suppress the dependence on P and just write $\mathcal{Q}$.

Response-Types. Next, I introduce the notion of response-types, which is central to the remaining analysis. Define the (random) vector

$$\mathbf{T} := \mathbf{T}(\mathbf{V}) := (g_D(z, \mathbf{V}) : z \in \mathcal{Z}) = (D_z : z \in \mathcal{Z})$$

which denotes an individual’s *response-type*.

Let $\mathcal{T} := \text{supp}(\mathbf{T}) = \{\mathbf{T}(\mathbf{v}) : \mathbf{v} \in \text{supp}(\mathbf{V})\}$ denote the set of possible response-types. We can view $\mathcal{T}$ as the set of possible functions from instruments to treatments, $\tau : \mathcal{Z} \rightarrow \mathcal{D}$. For each response-type τ , denote by $\tau(z)$ the treatment taken by those of response-type τ when $Z = z$.

$\mathcal{T}$ depends on the assumptions imposed on g_D and $\mathbf{V}$, and modulates the relationship between the observables and the unobservables. Indeed, Theorem T-1 in Heckman and Pinto (2018) implies that, for each treatment d and each outcome level y ,

$$\sum_{\tau \in \mathcal{T}} \mathbb{1}\{\tau(Z) = d\} P(\mathbf{T} = \tau) P(Y_d \leq y \mid \mathbf{T} = \tau) = P(D = d \mid Z) P(Y \leq y \mid D = d, Z) ,$$

almost surely.

¹⁰For completeness, I formally state and prove the equivalence of the two models in Appendix E.

If we do not place any restrictions on treatment selection, then $\mathcal{T} = \mathcal{D}^{\mathcal{Z}}$. Such assumptions can also restrict the distribution of $\mathbf{T}$ in other ways, beyond just restricting its support. Importantly, I show below that assumptions on g_D and/or $\mathbf{V}$ have identifying power *only* through restrictions on the distribution of $\mathbf{T}$. The distribution of $\mathbf{T}$ is captured by the (unknown) vector of response-type probabilities, which we denote by $\boldsymbol{\pi}$,

$$\boldsymbol{\pi} := \left(P(\mathbf{T} = \boldsymbol{\tau}) : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} \right).$$

Denote by $\boldsymbol{\pi}_Q$ the response-type probability vector induced by a particular parent distribution Q , i.e. $\boldsymbol{\pi}_Q := (Q(\mathbf{T} = \boldsymbol{\tau}) : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}})$.

Remark 2.1 (Extrapolation). As noted by Heckman and Pinto (2018), response-types define a coarse partition of the support of $\mathbf{V}$:

$$\{\mathbf{v} \in \mathbf{T}^{-1}(\boldsymbol{\tau})\} = \{\mathbf{v} \in \text{supp}(\mathbf{V}) \mid \mathbf{T}(\mathbf{v}) = \boldsymbol{\tau}\}.$$

So, we can define a generalized Marginal Treatment Response (MTR) function based on $\mathbf{V}$,

$$MTR_d(\mathbf{v}) := E(Y_d \mid \mathbf{V} = \mathbf{v}),$$

and get that $E(Y_d \mid \mathbf{T} = \boldsymbol{\tau}) = \frac{1}{P(\mathbf{V} \in \mathbf{T}^{-1}(\boldsymbol{\tau}))} \int_{\mathbf{T}^{-1}(\boldsymbol{\tau})} MTR_d(\mathbf{v}) dF_{\mathbf{V}}(\mathbf{v})$.

For simplicity, I focus on identification for counterfactual means and treatment effects conditional on response-types, and take the response-type support $\mathcal{T}$ to be a set. The framework here can accommodate more general parameters if we allow $\mathcal{T}$ to be a multiset (i.e. allow “redundant” response-types) with types that differ in ways other than the counterfactual choices made under the observed set of instrument values. For instance, if we are interested in a parameter of the form $E(Y_d \mid \mathbf{V} \in \mathcal{V}^*)$, for some measurable subset $\mathcal{V}^*$, we can define response-types based on the possible values for the vector $((g_D(z, \mathbf{V}) : z \in \mathcal{Z}), \mathbb{1}\{\mathbf{V} \in \mathcal{V}^*\})$. The remainder of the analysis proceeds in the same way,¹¹ but otherwise “identical” response-types can be treated differently by the treatment selection model (formally defined below, in Section 3) specified by the researcher. ♦

2.2. Motivating Examples. In this section, I introduce some simple motivating examples that fall into the general framework proposed in this paper.¹²

Example 1 (Head Start Impact Study). The Head Start Impact Study (HSIS) is a widely studied randomized evaluation of Head Start, a public childhood education program (see, e.g., Ludwig and Phillips, 2007; Gibbs, Ludwig, and Miller, 2011; Kline and Walters, 2016).

Let D denote preschool choice. Families choose between no preschool (n), competing preschools (c), and Head Start (h) for their child. HSIS randomized applicants into either being offered admission to Head Start ($Z = 1$), or not ($Z = 0$).

So, $\mathcal{D} := \{n, c, h\}$, $\mathcal{Z} := \{0, 1\}$. Assuming that, for all individuals, either $D_1 = D_0$ or $\mathbb{1}\{D_1 = h\} > \mathbb{1}\{D_0 = h\}$, we have that $\mathcal{T} := \{(n, n), (c, c), (h, h), (n, h), (c, h)\}$.

Shifting the instrument $Z : 0 \rightarrow 1$ causes two types of enrolment changes: $D : c \rightarrow h$ for those of type $\mathbf{T} = (c, h)$, and $D : n \rightarrow h$ for those of type $\mathbf{T} = (n, h)$.

¹¹Note that, for any particular $d \in \mathcal{D}$, the response-matrix $\mathbf{R}_d$ (defined below, in Section 3.2) may have redundant columns even when $\mathcal{T}$ has no redundant elements, and so, much of the analysis is not affected by the addition of more redundant columns.

¹²The results in this paper can be used to derive analytic bounds on each of the parameters considered in these examples, but I do not formally pursue this in this draft.

The following weighted average can be (point) identified

$$\begin{aligned} & \omega_{n \rightarrow h} E[Y_h - Y_n \mid D_0 = n, D_1 = h] + \omega_{c \rightarrow h} E[Y_h - Y_c \mid D_0 = c, D_1 = h] \\ &= \omega_{n \rightarrow h} E[Y_h - Y_n \mid \mathbf{T} = (n, h)] + \omega_{c \rightarrow h} E[Y_h - Y_c \mid \mathbf{T} = (c, h)] , \end{aligned}$$

where $\omega_{n \rightarrow h}$ and $\omega_{c \rightarrow h}$ are (point identified) weights, but the two component effects are generally only partially identified. $\blacklozenge$

Example 2 (Moving to Opportunity). “Moving to Opportunity” (MTO) is a widely studied randomized housing-mobility experiment (see, e.g., Katz, Kling, and Liebman, 2000; Chetty, Hendren, and Katz, 2015; Pinto, 2021).

Let D denote neighbourhood choice. Here, individuals choose from three different types of neighbourhoods: “high” poverty (H), “medium” poverty (M), and “low” poverty (L). Let $Z = (Z_M, Z_L)$ be the randomly assigned instrument, where Z_M and Z_L are binary, with Z_M indicating receipt of a voucher for M neighbourhoods, and Z_L indicating receipt of a voucher for L neighbourhoods. Individuals living in high poverty neighbourhoods were randomized into one of three groups:

- $(Z_M, Z_L) = (0, 0)$: Control group, received no vouchers,
- $(Z_M, Z_L) = (0, 1)$: Experimental group, received vouchers for L neighbourhoods only,
- $(Z_M, Z_L) = (1, 1)$: Section 8 group, received vouchers for both M and L neighbourhoods.

So, $\mathcal{D} := \{H, M, L\}$ and $\mathcal{Z} := \{(0, 0), (0, 1), (1, 1)\}$. Suppose we impose the “Unordered Partial Monotonicity” assumption of Mountjoy (2022), which requires that the following conditions hold for all individuals:

- (i) For all $z' \in \{0, 1\}$, (a) $\mathbb{1}\{D_{(0,z')} = M\} \leq \mathbb{1}\{D_{(1,z')} = M\}$ and (b) for all $d \in \mathcal{D} \setminus M$, $\mathbb{1}\{D_{(0,z')} = d\} \geq \mathbb{1}\{D_{(1,z')} = d\}$
- (ii) For all $z' \in \{0, 1\}$, (a) $\mathbb{1}\{D_{(z',0)} = L\} \leq \mathbb{1}\{D_{(z',1)} = L\}$ and (b) for all $d \in \mathcal{D} \setminus L$, $\mathbb{1}\{D_{(z',0)} = d\} \geq \mathbb{1}\{D_{(z',1)} = d\}$

Under these conditions, we have

$$\mathcal{T} := \left\{ \begin{array}{l} (H, H, H), (H, H, M), (H, L, M), (H, L, L), (M, M, M), \\ (M, L, M), (M, L, L), (L, L, M), (L, L, L) \end{array} \right\} .$$

Shifting $Z : (1, 1) \rightarrow (0, 1)$ causes two types of moves:

- Individuals with $\mathbf{T} \in \{(H, L, M), (M, L, M), (L, L, M)\}$: $M \rightarrow L$
- Individuals with $\mathbf{T} = (H, H, M)$: $M \rightarrow H$

The following weighted average can be (point) identified

$$\begin{aligned} & \omega_{M \rightarrow L} E[Y_L - Y_M \mid D_{(0,1)} = L, D_{(1,1)} = M] \\ & \quad + \omega_{M \rightarrow H} E[Y_H - Y_M \mid D_{(0,1)} = H, D_{(1,1)} = M] \\ &= \omega_{M \rightarrow L} E[Y_L - Y_M \mid \mathbf{T} \in \{(H, L, M), (M, L, M), (L, L, M)\}] \\ & \quad + \omega_{M \rightarrow H} E[Y_H - Y_M \mid \mathbf{T} = (H, H, M)] , \end{aligned}$$

where $\omega_{M \rightarrow L}$ and $\omega_{M \rightarrow H}$ are (point identified) weights, but the two component effects are generally only partially identified. $\blacklozenge$

Example 3 (Multi-dimensional Treatment). Consider a setting with a “primary” treatment, $P \in \{0, 1\}$, and a “supplementary” treatment, $S \in \{0, 1\}$, and define $D := (P, S)$.

Suppose some individuals are randomly offered incentives to participate in treatment. Let $Z = (Z_P, Z_S)$ where $Z_P \in \{0, 1\}$ represents an incentive to participate in P , and $Z_S \in \{0, 1\}$ represents an additional incentive to participate in S . Here, we have

$$\mathbf{T} := (D_{(0,0)}, D_{(1,0)}, D_{(1,1)}) =: ((P_{(0,0)}, S_{(0,0)}), (P_{(1,0)}, S_{(1,0)}), (P_{(1,1)}, S_{(1,1)})) .$$

Suppose individuals are offered the supplementary incentive only if they are offered the primary incentive (i.e. no individuals are offered only the supplementary incentive), and are only allowed to take the supplementary treatment if they also take the primary treatment. Individuals are always allowed, however, to choose among the three treatments, $\{(0, 0), (1, 0), (1, 1)\}$, regardless of the incentives they have been offered.

So, $\mathcal{D} := \{(0, 0), (1, 0), (1, 1)\}$ and $\mathcal{Z} := \{(0, 0), (1, 0), (1, 1)\}$. Assume that the following holds for all individuals:

- (i) For all $z' \in \{0, 1\}$, (a) $\mathbb{1}\{D_{(0,z')} \in \{(1, 0), (1, 1)\}\} \leq \mathbb{1}\{D_{(1,z')} \in \{(1, 0), (1, 1)\}\}$, and (b) $\mathbb{1}\{S_{(0,z')} = 1\} \leq \mathbb{1}\{S_{(1,z')} = 1\}$,
- (ii) For all $z' \in \{0, 1\}$, (a) $\mathbb{1}\{D_{(z',0)} = (1, 1)\} \leq \mathbb{1}\{D_{(z',1)} = (1, 1)\}$, and (b) $\mathbb{1}\{P_{(z',0)} = 1\} \leq \mathbb{1}\{P_{(z',1)} = 1\}$.

Under these conditions,

$$\mathcal{T} := \left\{ \begin{array}{l} ((0, 0), (0, 0), (0, 0)), ((0, 0), (0, 0), (1, 1)), ((0, 0), (1, 0), (1, 0)), \\ ((0, 0), (1, 0), (1, 1)), ((0, 0), (1, 1), (1, 1)), ((1, 0), (1, 0), (1, 0)), \\ ((1, 0), (1, 0), (1, 1)), ((1, 0), (1, 1), (1, 1)), ((1, 1), (1, 1), (1, 1)) \end{array} \right\} .$$

Many potential parameters of interest are only partially identified here. For example, in the decomposition,

$$\begin{aligned} & \underbrace{E[Y_{(1,1)} - Y_{(0,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{“Total” Effect}} \\ &= \underbrace{E[Y_{(1,1)} - Y_{(1,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{“Supplementary” Effect}} \\ & \quad + \underbrace{E[Y_{(1,0)} - Y_{(0,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{“Primary” Effect}} , \end{aligned}$$

all three effects are generally only partially identified. ♦

3. Main Results: Identification Theory for Treatment Selection Models

To make progress, I first formalize the notion of “treatment selection model”.

Definition 3.1 (Treatment Selection Model). A *treatment selection model* is a 3-tuple $\mathbb{S} = (\mathcal{V}_{\mathbb{S}}, \mathcal{F}_{\mathbb{S}}, \mathcal{G}_{\mathbb{S}})$, where $\mathcal{V}_{\mathbb{S}}$ represents restrictions on $\text{supp}(\mathbf{V})$, $\mathcal{F}_{\mathbb{S}}$ represents restrictions on the distribution of $\mathbf{V}$, and $\mathcal{G}_{\mathbb{S}}$ represents shape or functional form restrictions on g_D . Formally, $(\mathcal{V}_{\mathbb{S}}, \mathcal{F}_{\mathbb{S}}, \mathcal{G}_{\mathbb{S}})$ is such that

$\mathcal{V}_{\mathbb{S}}$ is a set ,

$\mathcal{F}_{\mathbb{S}}$ is a subset of the set of distributions $\{F \mid F \text{ is a distribution with } \text{supp}(F) \subseteq \mathcal{V}_{\mathbb{S}}\}$,

and $\mathcal{G}_{\mathbb{S}}$ is a subset of the set of functions $\{\mathcal{Z} \times \mathcal{V}_{\mathbb{S}} \rightarrow \mathcal{D}\}$.

A parent distribution $Q \in \mathcal{M}$ is said to satisfy treatment selection model $\mathbb{S}$ if and only if there exists random variables $(Y, D, Z, \mathbf{V}, \varepsilon_Y)$ and functions g_D, g_Y such that they satisfy Condition [M](#) and have $\text{Law}((g_Y(d, \mathbf{V}, \varepsilon_Y) : d \in \mathcal{D}), (g_D(z, \mathbf{V}) : z \in \mathcal{Z}), Z) = Q$, with $\text{supp}(\mathbf{V}) \subseteq \mathcal{V}_{\mathbb{S}}$, $\text{Law}(\mathbf{V}) \in \mathcal{F}_{\mathbb{S}}$ and $g_D \in \mathcal{G}_{\mathbb{S}}$.

Remark 3.1. Although the formal definition above takes treatment selection models to be specified as a 3-tuple, in practice we may often only impose assumptions on g_D (e.g., shape restrictions), leaving the distribution and support of $\mathbf{V}$ unrestricted. $\blacklozenge$

3.1. Structure of Identification in Treatment Selection Models. For any treatment selection model $\mathbb{S}$, define $\mathcal{Q}_{\mathbb{S}}$ as the identified set of parent distributions that satisfy $\mathbb{S}$,

$$\mathcal{Q}_{\mathbb{S}} := \{Q \in \mathcal{Q} \mid Q \text{ satisfies } \mathbb{S}\}.$$

Next, define $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ as the *a priori* restrictions imposed on the distribution of $\mathbf{T}$, represented by the vector response-type probabilities $\boldsymbol{\pi}$, i.e.

$$\begin{aligned} \mathcal{F}_{\mathbf{T}}(\mathbb{S}) &:= \left\{ \left(P_F \{ (g_D(z, \mathbf{V}) : z \in \mathcal{Z}) = \boldsymbol{\tau} \} : \boldsymbol{\tau} \in \mathcal{D}^{\mathcal{Z}} \right) \mid g_D \in \mathcal{G}_{\mathbb{S}}, F \in \mathcal{F}_{\mathbb{S}} \right\} \\ &= \{ \boldsymbol{\pi}_Q \mid Q \in \mathcal{M} \text{ and } Q \text{ satisfies } \mathbb{S} \}. \end{aligned}$$

$\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ is a feature of the treatment selection model $\mathbb{S}$ and does not depend on the observed data distribution P .¹³ Moreover, $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ can be represented as a subset of the set of all probability mass functions on $\mathcal{D}^{\mathcal{Z}}$ (i.e. the standard $\mathcal{D}^{\mathcal{Z}}$ -simplex, $\Delta^{\mathcal{D}^{\mathcal{Z}}}$), and is therefore always a finite-dimensional bounded set. We will refer to the set $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ as the *type restriction* associated with selection model $\mathbb{S}$.

As promised earlier, the following result shows that treatment selection models have identifying power only through their associated type restriction.

Theorem 3.1. For any treatment selection model $\mathbb{S}$,

$$\mathcal{Q}_{\mathbb{S}} = \{Q \in \mathcal{Q} \mid \boldsymbol{\pi}_Q \in \mathcal{F}_{\mathbf{T}}(\mathbb{S})\}.$$

Proof. See Appendix [A.1](#). $\blacksquare$

In other words, a parent distribution Q satisfies the treatment selection model $\mathbb{S}$ if and only if (i) Q is consistent with the observed data and witnesses Condition [M](#) (i.e. $Q \in \mathcal{Q}$) and (ii) the vector of response-type probabilities induced by Q is possible under treatment selection model $\mathbb{S}$.

¹³For example, consider the simple setting where the treatment and instrument are both binary, i.e. $|\mathcal{D}| = |\mathcal{Z}| = 2$, so that $\mathcal{D}^{\mathcal{Z}} = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. If $\mathbb{S}$ is the latent index threshold crossing model of Vytlačil (2002), then $\mathbb{S}$ simply imposes the restriction that $(1, 0) \notin \mathcal{T}$ (“no defiers”). In this case, $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ simply consists of all probability vectors, $\boldsymbol{\pi}$, on $\mathcal{D}^{\mathcal{Z}}$ that correspond to probability mass functions that assign 0 mass to the “defier” response-type, i.e. $\pi_{(1,0)} = 0$. This is a simple case – more generally, type restrictions can impose restrictions on the distribution of $\mathbf{T}$ beyond just restricting its support. For example, in the same setting, one can relax monotonicity to simply require that there are more “compliers” than “defiers”, in which case $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ will consist of all probability vectors on $\mathcal{D}^{\mathcal{Z}}$ such that $\pi_{(0,1)} > \pi_{(1,0)}$.

Remark 3.2. In general, given a data distribution P , in order to test whether $\mathbb{S}$ is consistent with the data, we need to check if $\mathcal{Q}_{\mathbb{S}}(P) \neq \emptyset$. One way to do this is to first find the identified set of parent distributions, $\mathcal{Q}(P)$, and then check if

$$\mathcal{Q}(P) \cap \{Q \mid Q \in \mathcal{M} \text{ and } Q \text{ satisfies } \mathbb{S}\} \neq \emptyset,$$

but, when $\mathcal{Y}$ is infinite, these two sets may be infinite-dimensional. Theorem 3.1 suggests that we can instead work with the identified set for π , $\Pi(P) := \{\pi_Q \mid Q \in \mathcal{Q}(P)\}$, and check if

$$\Pi(P) \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \neq \emptyset,$$

which is convenient because both $\Pi(P)$ and $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ are always finite-dimensional sets.

I emphasize, however, that the identified set for π , $\Pi(P)$, generally depends on the entire joint distribution (Y, D, Z) , and not just (D, Z) .¹⁴ The nature of this dependence is made explicit in Theorem 3.3, below. In fact, if $\Pi(P) \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \neq \emptyset$ then, for any $\pi^\dagger \in \Pi(P) \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S})$, we can construct a parent distribution Q such that $\pi_Q = \pi^\dagger$, $P_Q = P$, and $Q \in \mathcal{Q}_{\mathbb{S}}$. The proof of Theorem 3.3, provided in Appendix A.3, shows how to construct such a distribution Q . Although not explicitly stated in terms of response-type probabilities, the testable implications in Kitagawa (2021) are special cases of this approach (see Lemma A.1 in Kitagawa (2021)). ♦

Finally, I introduce some general notions, used in the remainder of the paper. Define $\mathcal{T}(\mathbb{S})$ as the set of all response-types not *a priori* eliminated by the selection model,

$$\begin{aligned} \mathcal{T}(\mathbb{S}) &:= \{\tau \in \mathcal{D}^Z \mid \pi \in \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \text{ such that } \pi_{(\tau)} > 0\} \\ &= \{\tau \in \mathcal{D}^Z \mid \exists Q \in \mathcal{Q}_{\mathbb{S}} \text{ such that } Q(\mathbf{T} = \tau) > 0\}, \end{aligned}$$

and note that $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ can always be represented as a subset of $\mathbb{R}^{|\mathcal{T}(\mathbb{S})|}$.

For any subset $\mathcal{T} \subseteq \mathcal{D}^Z$, define $\mathcal{Q}_{\mathcal{T}}$ as the identified set for parent distributions with response-type support in $\mathcal{T}$,

$$\mathcal{Q}_{\mathcal{T}} := \left\{ Q \in \mathcal{Q} \mid \sum_{\tau \in \mathcal{T}} Q(\mathbf{T} = \tau) = 1 \right\}.$$

In general, $\mathcal{Q}_{\mathbb{S}} \subseteq \mathcal{Q}_{\mathcal{T}(\mathbb{S})}$, but in the important special case where $\mathbb{S}$ simply eliminates some response-types, we have that $\mathcal{Q}_{\mathbb{S}} = \mathcal{Q}_{\mathcal{T}(\mathbb{S})}$.

Remark 3.3 (“Translating” a Selection Model into a Type Restriction). I will generally treat $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ as known, given $\mathbb{S}$. I emphasize, however, that “translating” such models from more “structural” representations can be non-trivial, in general. A well-known example of such translation is Vytlačil (2002, 2006).

Given some representation of $\mathbb{S}$, it is often much easier to derive $\mathcal{T}(\mathbb{S})$ than $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$. In cases where $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ is not known but $\mathcal{T}(\mathbb{S})$ is, this paper’s framework can be used to construct identified sets for parameters over $Q \in \mathcal{Q}_{\mathcal{T}(\mathbb{S})}$ to obtain valid bounds (potentially not sharp), useful for (potentially conservative) inference. ♦

The next section develops methodology for identifying response-type conditional counterfactual distributions and parameters.

¹⁴This has been noted, in the binary treatment and instrument setting, by Huber et al. (2017), and is also implicit in the analysis of Kitagawa (2021).

3.2. Identified Set for Parent Distributions. I introduce the following shorthand notation, used throughout the remainder. For each $Q \in \mathcal{M}$, define

$$\pi \xi_{Y_d, Q}(y) := \left(Q(T = \tau) f_{Q; Y_d}(y | T = \tau) : \tau \in \mathcal{D}^Z \right),$$

where $f_{Q; Y_d}(y | T = \tau)$ is the density associated with $Q(Y_d \leq y | T = \tau)$. Note that $\int_{\mathcal{Y}} \pi \xi_{Y_d, Q}(y) d\mu(y) = \pi_Q$.

When the response-type support $\mathcal{T}$ is clear, we will treat π_Q and $\pi \xi_{Y_d, Q}(y)$ as $|\mathcal{T}|$ -vectors (instead of $|\mathcal{D}^Z|$ -vectors), a slight abuse of notation. We will often suppress the dependence on Q and just write π and $\pi \xi_{Y_d}(y)$.

For any $\mathcal{T} \subseteq \mathcal{D}^Z$, let $\mathcal{T} =: \{\tau_1, \dots, \tau_{|\mathcal{T}|}\}$ be an ordering of response-types, and let $\mathcal{Z} =: \{z_1, \dots, z_{|\mathcal{Z}|}\}$ be some ordering of the instrument levels; throughout, assume all relevant tuples follow this ordering.

Next, define, for each $d \in \mathcal{D}$, the $|\mathcal{Z}| \times |\mathcal{T}|$ matrix $\mathbf{R}_{d, \mathcal{T}}$ as the binary matrix such that its (i, j) -th component is $\mathbb{1}\{\tau_j(z_i) = d\}$. Note that $\mathbf{R}_{d, \mathcal{T}}$ is feature of the model itself: it is known from $\mathcal{T}$, and does not depend on the observed data. When the $\mathcal{T}$ of interest is clear, we will often suppress the dependence on $\mathcal{T}$ and just write $\mathbf{R}_d$ ¹⁵.

Finally, define $\mathbf{pf}_d$ as the (observed) vector of propensity-score-weighted densities,

$$\mathbf{pf}_d(y) := \left(P(D = d | Z = z) f_{P; Y}(y | D = d, Z = z) : z \in \mathcal{Z} \right),$$

where $f_{P; Y}(y | D = d, Z = z)$ is the density associated with $P(Y \leq y | D = d, Z = z)$. Note that $\mathbf{pf}_d$ is identified since it is a function of the observed data distribution P .

The following theorem characterizes the role of $\mathcal{T}$ in modulating the relationship between the observed data distribution P and the identified set of parent distributions.

Theorem 3.2 (Sharp Characterization of $\mathcal{Q}_{\mathcal{T}}$). *Suppose P satisfies Condition D. Then, for any $\mathcal{T} \subseteq \mathcal{D}^Z$, $Q \in \mathcal{Q}_{\mathcal{T}}$ if and only if $Q \in \mathcal{M}$, $Q_Z = P_Z$ and, for each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, the following equality holds*

$$\mathbf{R}_d \pi \xi_{Y_d}(y) = \mathbf{pf}_d(y). \quad (\star)$$

Proof. See Appendix A.2. ■

As noted earlier, the $\implies$ direction in Theorem 3.2 follows from Theorem T-1 in Heckman and Pinto (2018). The interesting part of this result is that it is sharp (i.e. the $\impliedby$ part of the above result), but it is not immediately operationalizable since (in general) it represents a continuum of equalities. Nevertheless, Theorem 3.2 suggests taking $\pi \xi_{Y_d, Q}$, for $d \in \mathcal{D}$, as the primitives in our model. In other words, this result suggests that the key ingredients for identification analysis in this setting are the distributions (Y_d, T) , for $d \in \mathcal{D}$.

Remark 3.4. For any $d \in \mathcal{D}$, integrating both sides of $(\star)$ over $y \in \mathcal{Y}$, generates the propensity score equality:

$$\mathbf{R}_d \pi = \left(P(D = d | Z = z) : z \in \mathcal{Z} \right). \quad (3)$$

If Y is observed for all treatment states, as is the case here, the propensity score equality is redundant, given $(\star)$. In some settings, however, (Y, D, Z) may only be observed conditional on $D \in \mathcal{D}^\dagger$, where $\mathcal{D}^\dagger \subsetneq \mathcal{D}$, with only (D, Z) being observed when $D \in \mathcal{D} \setminus \mathcal{D}^\dagger$ (i.e. there is

¹⁵Heckman and Pinto (2018) denote this matrix $\mathbf{B}_d$ in their notation.

sample selection). In such cases, Theorem 3.2 continues to hold if, for each $d \in \mathcal{D} \setminus \mathcal{D}^\dagger$, the requirement that $(\star)$ must hold for each $y \in \mathcal{Y}$ is replaced with the (much simpler) requirement that (3) must hold.¹⁶ With this modification in place, the approach of this paper extends to settings with sample selection. For an example of such an application, see Kroft, Mourifié, and Vayalinal (2024). $\blacklozenge$

Remark 3.5. If $\mathcal{Y}$ is finite, then we can use Theorem 3.2 to set up our identification problem as one large linear programming problem and proceed using, e.g. Fang, Santos, Shaikh, and Torgovitsky (2023) or Cho and Russell (2023) for inference. I do not pursue this approach here. The main focus of this paper is on an alternate approach that can accommodate both finite and continuous outcomes. This approach is also less “black box” than purely computational approaches, allowing for the symbolic derivation of analytic expressions for bounds on target parameters, providing insights into the source and structure of identifying power in the types of models considered here. $\blacklozenge$

3.3. Identified Set for Response-Type Probabilities and Mean Counterfactuals. In this subsection, we take $\mathcal{T} \subseteq \mathcal{D}^\mathbb{Z}$ as given. Define $\Pi_{\mathcal{T}}$ as the identified set for π when the response-type support is restricted to $\mathcal{T}$,

$$\Pi_{\mathcal{T}} := \{\pi_Q : Q \in \mathcal{Q}_{\mathcal{T}}\},$$

which we will generally view as a subset of $\mathbb{R}^{|\mathcal{T}|}$.

Before proceeding, we define what it means for a set to be a convex polytope. This and related notions are used throughout the remainder of the paper.

Definition 3.2. A set $K \subseteq \mathbb{R}^n$ is a **convex polytope** if and only if there exists, for some $m \in \mathbb{N}$, a matrix $A \in \mathbb{R}^{m \times n}$ and vector $b \in \mathbb{R}^m$ such that

$$K = \{x \in \mathbb{R}^n \mid Ax \leq b\}.$$

If K is a bounded convex polytope, then K is the convex hull of a finite set in $\mathbb{R}^n$.

Fix some $d \in \mathcal{D}$. For any $y \in \mathcal{Y}$, let $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \subseteq \mathbb{R}_{\geq 0}^{|\mathcal{T}|}$ be the (potentially empty or unbounded) convex polytope defined by the set of non-negative solutions to $(\star)$ at (d, y) ,

$$\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) := \left\{ \mathbf{x} \in \mathbb{R}_{\geq 0}^{|\mathcal{T}|} \mid \mathbf{R}_d \mathbf{x} = \mathbf{p}\mathbf{f}_d(y) \right\}.$$

$\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))$ is identified since it is a function of $\mathbf{p}\mathbf{f}_d$, which depends only on the observed data distribution P .

Next, define $\Gamma_d(\mathcal{T})$ as the set of non-negative measurable vector-valued functions $\mathbf{g} : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{|\mathcal{T}|}$ such that $\mathbf{g}(y) \in \mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))$ for each $y \in \mathcal{Y}$,

$$\Gamma_d(\mathcal{T}) := \left\{ \mathbf{g} \in \left\{ \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{|\mathcal{T}|} \right\} \mid \begin{array}{l} \mathbf{g} \text{ is measurable,} \\ \mathbf{g}(y) \in \mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \forall y \in \mathcal{Y} \end{array} \right\}.$$

We will generally suppress the dependence of Γ_d on $\mathcal{T}$. Note that if $\mathcal{T}$ is such that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) = \emptyset$ for some $y \in \mathcal{Y}$, then $\Gamma_d = \emptyset$. Theorem 3.2 implies that $\pi \xi_{Y_d}$ satisfies $(\star)$ for every $y \in \mathcal{Y}$, and so, we must have that $\pi \xi_{Y_d} \in \Gamma_d$.

¹⁶We also know that $\text{supp}_Q(Y_d) = \mathcal{Y}$, but this is automatically satisfied by any $Q \in \mathcal{M}$.

Now consider the set, denoted $\Pi_{\mathcal{T}|d}$, formed by the integrals of functions in Γ_d ,

$$\Pi_{\mathcal{T}|d} := \left\{ \int_{\mathcal{Y}} g(y) d\mu(y) \mid g \in \Gamma_d \right\}.$$

Clearly, $\pi \xi_{Y_d} \in \Gamma_d$ implies that $\pi \in \Pi_{\mathcal{T}|d}$. Of course, if $\mathcal{B}_d(\mathbf{p}f_d(y)) = \emptyset$ for some $y \in \mathcal{Y}$, then $\Pi_{\mathcal{T}|d} = \emptyset$. By linearity of integration, we have that $\Pi_{\mathcal{T}|d}$ is a closed convex set and so, whenever it is non-empty, we will work with its *support function*, defined as follows.

Definition 3.3. For any non-empty closed convex set $C \subseteq \mathbb{R}^n$, let $h_C : \mathbb{R}^n \rightarrow \mathbb{R}$ be its **support function**, which is defined as the convex function such that, for every $\mathbf{c} \in \mathbb{R}^n$,

$$h_C(\mathbf{c}) := \sup_{\mathbf{x} \in C} \mathbf{c}^\top \mathbf{x}.$$

Every non-empty closed convex set is uniquely determined by its support function.

Whenever $\mathcal{B}_d(\mathbf{p}f_d(y)) \neq \emptyset$, denote the support function of $\mathcal{B}_d(\mathbf{p}f_d(y))$ as $h_{\mathbf{R}_d}(\cdot; \mathbf{p}f_d(y))$,

$$h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}f_d(y)) := h_{\mathcal{B}_d(\mathbf{p}f_d(y))}(\mathbf{c}) = \sup_{\mathbf{x} \geq 0: \mathbf{R}_d \mathbf{x} = \mathbf{p}f_d(y)} \mathbf{c}^\top \mathbf{x}.$$

In Section 5, I discuss the computation and estimation of the function $h_{\mathbf{R}_d}$, which is an important object in our analysis. The next result relates $h_{\mathbf{R}_d}$ to the support function of $\Pi_{\mathcal{T}|d}$: the boundary of $\Pi_{\mathcal{T}|d}$ can be obtained by integrating along the boundary of $\mathcal{B}_d(\mathbf{p}f_d(\cdot))$.

Lemma 3.1. Suppose P satisfies Condition D. Then, if $\mathcal{T} \subseteq \mathcal{D}^{\mathbb{Z}}$ is such that $\mathcal{B}_d(\mathbf{p}f_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$, then, for all $\mathbf{c} \in \mathbb{R}^{|\mathcal{T}|}$

$$h_{\Pi_{\mathcal{T}|d}}(\mathbf{c}) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}f_d(y)) d\mu(y).$$

Further, $\Pi_{\mathcal{T}|d}$ is a convex polytope.

Proof. Follows from suitable applications of Proposition 1.2 and Theorem 1.3 in Billera and Sturmfels (1992), see Appendix A.4. ■

Finally, the argument that $\pi \in \Pi_{\mathcal{T}|d}$ holds for each $d \in \mathcal{D}$, and so, we must have $\pi \in \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d}$. The next result shows that this is sharp.

Theorem 3.3. Suppose P satisfies Condition D. Then, for any $\mathcal{T} \subseteq \mathcal{D}^{\mathbb{Z}}$,

$$\Pi_{\mathcal{T}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d},$$

and $\Pi_{\mathcal{T}}$ is a compact convex polytope in $\mathbb{R}_{\geq 0}^{|\mathcal{T}|}$. Moreover, $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$ if and only if $\Pi_{\mathcal{T}} \neq \emptyset$.

Proof. See Appendix A.3. ■

In Theorem 3.3, the condition $\Pi_{\mathcal{T}} \neq \emptyset$ can be viewed as first requiring that, for each $d \in \mathcal{D}$, $\mathcal{B}_d(\mathbf{p}f_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$ (so that $\Pi_{\mathcal{T}|d} \neq \emptyset$) and then requiring that the $\{\Pi_{\mathcal{T}|d}\}_{d \in \mathcal{D}}$ have non-empty intersection. Although the second condition automatically implies the first, this distinction will be relevant for practical implementation.

Theorem 3.3 immediately implies the following general results.

Corollary 3.1 (Identified Set for Response-Type Probabilities). *Suppose P satisfies Condition D. For any treatment selection model $\mathbb{S}$, define the identified set for response-type probabilities under $\mathbb{S}$ as $\Pi_{\mathbb{S}} := \{\pi_Q : Q \in \mathcal{Q}_{\mathbb{S}}\}$. Then,*

$$\Pi_{\mathbb{S}} := \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \cap \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d},$$

where $\Pi_{\mathbb{S}}$ and $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ are represented as subsets of $\mathbb{R}^{|\mathcal{T}(\mathbb{S})|}$. Moreover, if $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ is a convex polytope, then $\Pi_{\mathbb{S}}$ is a convex polytope.

Proof. The first claim follows immediately from Theorems 3.1 and 3.3. The second claim follows from Theorem 3.3 by noting that the intersection of convex polytopes is always a convex polytope. ■

Corollary 3.2 (Sharp Testable Implications of Treatment Selection Models). *Suppose P satisfies Condition D. Then,*

$$\mathcal{Q}_{\mathbb{S}} \neq \emptyset \iff \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \cap \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \neq \emptyset,$$

where $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ is represented as a subset of $\mathbb{R}^{|\mathcal{T}(\mathbb{S})|}$.

Proof. Follows immediately from Theorem 3.1 and Corollary 3.1. ■

Remark 3.6 (Sharp Test for Random Assignment). Theorem 3.3 does not assume that Condition M holds (i.e. that $\mathcal{Q} \neq \emptyset$); indeed, setting $\mathcal{T} = \mathcal{D}^Z$ in Theorem 3.3 yields (sharp) testable implications for $\mathcal{Q}_{\mathcal{D}^Z} = \mathcal{Q} \neq \emptyset$ (i.e. for Condition M). If we assume that Condition M(i) holds, then this represents a test for random assignment of Z . ♦

Remark 3.7 (Connection to Random Set Theory). Observe that if $\mathcal{Y}$ is finite, then $\Pi_{\mathcal{T}|d}$ is just the Minkowski sum of the convex polytopes $\{\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))\}_{y \in \mathcal{Y}}$. More generally, $\Pi_{\mathcal{T}|d}$ can be viewed as the *Minkowski integral* of the polytopes $\{\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))\}_{y \in \mathcal{Y}}$, where the notion of Minkowski integral used here is the continuous analogue to the Minkowski sum introduced and formalized in Billera and Sturmfels (1992).¹⁷

The Minkowski integral bears a strong resemblance to the notion of Aumann (or selection) expectation from random set theory, as formalized by Artstein and Vitale (1975). Indeed, both the Minkowski integral and the Aumann expectation are adaptations of the integral for set-valued functions introduced by Aumann (1965), and it is well-known that results analogous to Lemma 3.1 are available for the Aumann expectation as well (see, e.g., Theorem 3.11 in Molchanov and Molinari (2018)). Informally, the random set theory literature applies this integral to a random set (i.e. a measurable correspondence over the sample space¹⁸) such that the *realizations* of the random variable of interest always lies within the corresponding realizations of this set, whereas the approach in this paper applies this integral to a

¹⁷This is an informal version of the argument used to prove Lemma 3.1, which proceeds by viewing the correspondence $y \mapsto \mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))$ as (a generalized version of) a *polytope bundle* (Billera and Sturmfels, 1992), and leveraging properties of Minkowski integrals of polytope bundles; see Appendix A.4 for details.

¹⁸For formal definitions, see (e.g.) Molchanov (2017) or Molchanov and Molinari (2018).

(non-stochastic) correspondence such that the graph of the *distribution* (density) of the random variable of interest lies inside the graph of this correspondence; the difference in the two approaches can be viewed as analogous to the distinction between $\int_{\Omega} X(\omega) dP(\omega)$ and $\int_{\mathbb{R}} xf_X(x) dx$ as representations for the expectation $E[X]$ of a real-valued random variable X with density function f_X . ♦

Remark 3.8. We have that Γ_d , which only depends on $\mathbb{S}$ through $\mathcal{T}(\mathbb{S})$, represents pointwise restrictions on $\pi\xi_{Y_d}$; informally, this can be interpreted as the “direct” restrictions imposed on the distribution $(Y_d, \mathbf{T})$. For each $d \in \mathcal{D}$, these direct restrictions on $(Y_d, \mathbf{T})$, in turn, impose restrictions on the distribution of $\mathbf{T}$, and these restrictions must all be compatible with each other and also with any additional restrictions imposed by $\mathbb{S}$.

The “direct” restrictions on $(Y_d, \mathbf{T})$ can be viewed as restricting both the conditional distribution $Y_d|\mathbf{T}$ and the *relationship* between the distribution of $Y_d|\mathbf{T}$ and the distribution of $\mathbf{T}$, and then the “compatibility” condition restricts the distribution of $\mathbf{T}$ which, in turn, restricts the distribution of $Y_d|\mathbf{T}$ only indirectly via the restricted relationship between the distributions of $Y_d|\mathbf{T}$ and $\mathbf{T}$.

In particular, the set of allowed response-types directly restricts the distributions of $Y_d|\mathbf{T}$ while any other restrictions on the distribution of $\mathbf{T}$ only restricts “how much” mass is available to allocate to $(Y_d, \mathbf{T})$ but not “where” (in $\mathcal{Y}$) the available mass can be allocated. This is intuitively similar to Horowitz and Manski (1995)’s seminal result on the partial identification of mixture components (in the context of contaminated data).

I emphasize, however, that these “indirect” restrictions can nevertheless provide significant identification power in many cases, which is somewhat different from simpler settings (e.g., binary Z and D). For an extreme case, see Remark 3.10 which outlines knife-edge conditions under which indirect restrictions can deliver point-identification. ♦

As a by-product of the proof of Theorem 3.3, we have the following corollary, which also helps highlight the intuition discussed above.

Corollary 3.3 (Identified Set for Marginal Outcome Distributions). *Suppose P satisfies Condition D. Then, for any treatment selection model $\mathbb{S}$, we have that the identified set for $\pi\xi_{Y_d}$ under $\mathbb{S}$ is given by*

$$\{\pi\xi_{Y_d, Q} \mid Q \in \mathcal{Q}_{\mathbb{S}}\} = \left\{ \mathbf{g} \in \Gamma_d(\mathcal{T}(\mathbb{S})) \mid \int_{\mathcal{Y}} \mathbf{g}(y) d\mu(y) \in \Pi_{\mathbb{S}} \right\}.$$

Further, the joint identification region¹⁹ for $(\pi\xi_{Y_d} : d \in \mathcal{D})$ under $\mathbb{S}$ is given by

$$\begin{aligned} & \{(\pi\xi_{Y_d, Q} : d \in \mathcal{D}) \mid Q \in \mathcal{Q}_{\mathbb{S}}\} \\ &= \left\{ (\mathbf{g}_d : d \in \mathcal{D}) \in \prod_{d \in \mathcal{D}} \Gamma_d(\mathcal{T}(\mathbb{S})) \mid \exists \pi \in \Pi_{\mathbb{S}} \text{ s.t. } \int_{\mathcal{Y}} \mathbf{g}_d(y) d\mu(y) = \pi \forall d \in \mathcal{D} \right\}. \end{aligned}$$

Proof. Follows from the proof of Theorem 3.3. See Appendix A.3 for details. ■

¹⁹Note that this is the joint identified region for the *marginal* densities, not the identified set for joint distributions (which is provided in Theorem 3.6, below). In particular, this result does not directly say anything about the dependence between different marginal outcomes distributions.

Corollary 3.3 can be applied to obtain sharp bounds on mean outcomes and treatment effects, as follows. Suppose we are interested in the identified set for $E(\kappa(Y_d) \mid \mathbf{T} \subseteq \mathcal{T}^*)$, where the function $\kappa : \mathcal{Y} \rightarrow \mathbb{R}$ and subset $\mathcal{T}^* \subseteq \mathcal{T}$ are known. Note that

$$E[\kappa(Y_d) \mid \mathbf{T} \subseteq \mathcal{T}^*] = \frac{\sum_{\tau \in \mathcal{T}^*} P(\mathbf{T} = \tau) E(\kappa(Y_d) \mid \mathbf{T} = \tau)}{\sum_{\tau \in \mathcal{T}^*} P(\mathbf{T} = \tau)}.$$

Corollary 3.3 suggests a representation for the identified set for parameters of this form. To proceed, first define the following sets, for any $\mathcal{T} \subseteq \mathcal{D}^Z$, $d \in \mathcal{D}$, and selection model $\mathbb{S}$,

$$\begin{aligned} \Theta_{\mathcal{T}|d}(\kappa) &:= \left\{ \left(\int_{\mathcal{Y}} \kappa(y) \mathbf{g}(y) d\mu(y), \int_{\mathcal{Y}} \mathbf{g}(y) d\mu(y) \right) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{T}|} \mid \mathbf{g} \in \Gamma_d(\mathcal{T}) \right\}, \\ \tilde{\Pi}_{\mathbb{S}} &:= \left\{ (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{T}|} \mid \boldsymbol{\theta}_d \in \kappa(\mathcal{Y})^{|\mathcal{T}|}, \boldsymbol{\pi} \in \Pi_{\mathbb{S}} \right\}. \end{aligned}$$

Note that the set $\tilde{\Pi}_{\mathbb{S}}$ is just the representation of $\Pi_{\mathbb{S}}$ in a higher dimensional space. $\Theta_{\mathcal{T}|d}$ is a closed convex set and, as with $\Pi_{\mathcal{T}|d}$, the support function of $\Theta_{\mathcal{T}|d}$ can be obtained in terms of $h_{\mathbf{R}_d}$, as follows.

Lemma 3.2. *Suppose P satisfies Condition D. Then, if $\mathcal{T} \subseteq \mathcal{D}^Z$ is such that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$, then, for all $(\mathbf{c}_1, \mathbf{c}_2) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{T}|}$,*

$$h_{\Theta_{\mathcal{T}|d}(\kappa)}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\kappa(y) \mathbf{c}_1 + \mathbf{c}_2; \mathbf{p}\mathbf{f}_d(y)) d\mu(y).$$

Proof. See Appendix A.4. ■

We are now ready to state the following identification result for mean counterfactual outcomes and treatment effects, as suggested by Corollary 3.3.

Theorem 3.4 (Identified Set for Mean Counterfactuals). *Suppose P satisfies Condition D. Let a treatment selection model $\mathbb{S}$, a function $\kappa : \mathcal{Y} \rightarrow \mathbb{R}$, and a subset $\mathcal{T}^* \subseteq \mathcal{T}(\mathbb{S})$ be given, and let*

$$\mathbf{c} := (\mathbf{1}\{\tau \in \mathcal{T}^*\} : \tau \in \mathcal{T}(\mathbb{S})) .$$

Then, for any $d \in \mathcal{D}$,

$$\begin{aligned} \left\{ Q(\mathbf{T} \in \mathcal{T}^*) E_Q[\kappa(Y_d) \mid \mathbf{T} \in \mathcal{T}^*] \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} &= \left\{ \mathbf{c}^\top \boldsymbol{\theta}_d \mid (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \right\}, \\ \text{and } \left\{ E_Q[\kappa(Y_d) \mid \mathbf{T} \in \mathcal{T}^*] \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} &= \left\{ \frac{\mathbf{c}^\top \boldsymbol{\theta}_d}{\mathbf{c}^\top \boldsymbol{\pi}} \mid (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \right\}. \end{aligned}$$

Further, for any $d \neq d' \in \mathcal{D}$,

$$\begin{aligned} \left\{ Q(\mathbf{T} \in \mathcal{T}^*) E_Q[\kappa(Y_d) - \kappa(Y_{d'}) \mid \mathbf{T} \in \mathcal{T}^*] \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} \\ = \left\{ \mathbf{c}^\top \boldsymbol{\theta}_d - \mathbf{c}^\top \boldsymbol{\theta}_{d'} \mid \begin{array}{l} (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}, \\ (\boldsymbol{\theta}_{d'}, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\}, \end{aligned}$$

and

$$\begin{aligned} \left\{ E_Q[\kappa(Y_d) - \kappa(Y_{d'}) \mid \mathbf{T} \in \mathcal{T}^*] \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} \\ = \left\{ \frac{\mathbf{c}^\top \boldsymbol{\theta}_d - \mathbf{c}^\top \boldsymbol{\theta}_{d'}}{\mathbf{c}^\top \boldsymbol{\pi}} \mid \begin{array}{l} (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}, \\ (\boldsymbol{\theta}_{d'}, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\}. \end{aligned}$$

Proof. Follows from the proof of Theorem 3.3, via Corollary 3.3. See Appendix A.3. ■

Remark 3.9. For simplicity, Theorem 3.4 only reports the case where c is a binary vector and only considers treatment effects involving two identically weighted outcomes, but the analogous result applies to any weighted average of response-type conditional outcome means.

Moreover, note that additional assumptions on the response-type conditional means can be imposed by adding suitable conditions to the definition of $\tilde{\Pi}_{\mathbb{S}}$. This includes assumptions like $E(Y_d \mathbb{1}\{T = \tau\}) \geq E(Y_{d'} \mathbb{1}\{T = \tau\})$, or (discrete analogues to) the “comparable compliers” assumption of Mountjoy (2022), or more complex linear relationships. Non-linear restrictions can also be imposed, but this will complicate the computation of the support function of $\tilde{\Pi}_{\mathbb{S}}$, required for the inference approach proposed in this paper (see Section 4). ♦

3.3.1. *Point Identification.* For completeness, I also revisit sufficient conditions for point identification, due to Heckman and Pinto (2018).

Proposition 3.5 (Heckman and Pinto (2018)). *Suppose P satisfies Condition D. Then, for any treatment selection model $\mathbb{S}$ with $\mathcal{Q}_{\mathbb{S}} \neq \emptyset$, the following statements hold, under $\mathbb{S}$:*

- (i) *If $c \in \text{row-span}(\mathbf{R}_{d, \mathcal{T}(\mathbb{S})})$, then $c^\top \pi \xi_{Y_d}$ is point-identified*
- (ii) *If $c \in \text{row-span}\left(\begin{bmatrix} \mathbf{R}_{1, \mathcal{T}(\mathbb{S})} \\ \vdots \\ \mathbf{R}_{N_D, \mathcal{T}(\mathbb{S})} \end{bmatrix}\right)$, then $c^\top \pi$ is point-identified*

Proposition 3.5 is mathematically equivalent to Theorem T-2 in Heckman and Pinto (2018), where the result is stated in terms of Moore-Penrose inverses of $\mathbf{R}_d$. Note that Proposition 3.5(i) is also sufficient for the point identification of mean outcomes, since $c \in \text{row-span}(\mathbf{R}_{d, \mathcal{T}(\mathbb{S})})$ implies that $\{c^\top \theta_d \mid (\theta_d, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}\}$ and $\{\frac{c^\top \theta_d}{c^\top \pi} \mid (\theta_d, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}\}$ are singletons. Similarly, for point identification of treatment effects, $c \in \text{row-span}(\mathbf{R}_{d, \mathcal{T}(\mathbb{S})}) \cap \text{row-span}(\mathbf{R}_{d', \mathcal{T}(\mathbb{S})})$ is a sufficient condition for $\left\{c^\top \theta_d - c^\top \theta_{d'} \mid \begin{array}{l} (\theta_d, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}, \\ (\theta_{d'}, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\}$ and $\left\{\frac{c^\top \theta_d - c^\top \theta_{d'}}{c^\top \pi} \mid \begin{array}{l} (\theta_d, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}}, \\ (\theta_{d'}, \pi) \in \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\}$ to collapse to a point.

These conditions are not, in general, *necessary* for point-identification. In the absence of these conditions, point identification can still be obtained if some “knife-edge” conditions hold, as discussed in the following remark.²⁰

Remark 3.10 (“Knife-edge” Conditions for Point Identification). Let $\mathbb{S}$ be a treatment selection model such that $\mathcal{Q}_{\mathbb{S}} \neq \emptyset$. If $\Pi_{\mathbb{S}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S})$ is a singleton, say $\{p\}$, then obviously $\pi = p$ is point-identified, but the (weighted) density $\pi \xi_{Y_d}$ may not be point-identified for any d . If p is a *vertex* of $\Pi_{\mathcal{T}(\mathbb{S})|d}$, however, then $\pi \xi_{Y_d}$ is point-identified! I sketch the argument below. A similar argument shows that $c^\top \pi \xi_{Y_d}$ is point-identified if $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S})$ is a (non-empty) subset of the extreme face of the polytope $\Pi_{\mathcal{T}(\mathbb{S})|d}$ with respect to the direction c .²¹

²⁰See also Navjeevan, Pinto, and Santos (2023) for necessary and sufficient conditions for point identification in a broader class of models, including those considered here.

²¹In other words, if $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \subseteq \arg\max_{x \in \Pi_{\mathcal{T}(\mathbb{S})|d}} c^\top x$.

To see this, we can use the fact that any vertex of $\Pi_{\mathcal{T}(\mathbb{S})|d}$ is the result of integrating along the corresponding vertex of $\mathcal{B}_d(\mathbf{p}f_d(y))$. Indeed, it is possible to show that the extreme face of the polytope $\Pi_{\mathcal{T}(\mathbb{S})|d}$, with respect to a direction $\mathbf{c}$, is the following set

$$\left\{ \int_{\mathcal{Y}} \mathbf{g}(y) d\mu(y) \mid \begin{array}{l} \mathbf{g} : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{|\mathcal{T}(\mathbb{S})|} \text{ is measurable,} \\ \mathbf{g}(y) \in \mathcal{B}_d^{\mathbf{c}}(\mathbf{p}f_d(y)) \forall y \in \mathcal{Y} \end{array} \right\}, \quad (4)$$

where $\mathcal{B}_d^{\mathbf{c}}(\mathbf{p}f_d(y))$ is defined as the extreme face of the polytope $\mathcal{B}_d(\mathbf{p}f_d(y))$ with respect to direction $\mathbf{c}$.²² Since $\mathbf{p}$ is a vertex of $\Pi_{\mathcal{T}(\mathbb{S})|d}$, there exists a direction $\mathbf{c}_{\mathbf{p}}$ such that $\operatorname{argmax}_{\mathbf{x} \in \Pi_{\mathcal{T}(\mathbb{S})|d}} \{\mathbf{c}_{\mathbf{p}}^T \mathbf{x}\} = \{\mathbf{p}\}$. Now, it is clear that (4) is a singleton if and only if $\mathcal{B}_d^{\mathbf{c}_{\mathbf{p}}}(y)$ is a singleton, say $\{\mathbf{b}_d^{\mathbf{c}_{\mathbf{p}}}(y)\}$, for every $y \in \mathcal{Y}$, which immediately gives us the result. Indeed, when this is the case, $\pi \xi_{Y_d}(y) = \mathbf{b}_d^{\mathbf{c}_{\mathbf{p}}}(y)$ for every $y \in \mathcal{Y}$. Of course, this also implies the point-identification of the corresponding mean counterfactuals. $\blacklozenge$

3.4. Identified Set for Joint Distribution of Counterfactual Outcomes. The next result shows that while treatment selection models can provide information about the *joint* distribution of potential outcomes, $(Y_d : d \in \mathcal{D})$, they do so only through restrictions on the response-type conditional *marginals*, $Y_d \mid \mathbf{T} = \tau$, and on the response-type probability vector.

Theorem 3.6. (*Identified Set for Joint Outcome Distribution*) Suppose P satisfies Condition *D*, and let a treatment selection model $\mathbb{S}$ be given. Then, for any treatment selection model $\mathbb{S}$, the identified set, under $\mathbb{S}$, for the conditional joint distributions of $(Y_d : d \in \mathcal{D})$ given $\mathbf{T}$ is as follows

$$\begin{aligned} & \left\{ \left(Q_{(Y_d : d \in \mathcal{D}) | \mathbf{T} = \tau} : \tau \in \mathcal{T}(\mathbb{S}) \right) \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} \\ &= \left\{ \left(C_{\tau} \left(\left(\mathfrak{F}[\mathbf{g}_{(d)(\tau)}] : d \in \mathcal{D} \right) \right) : \tau \in \mathcal{T}(\mathbb{S}) \right) \mid \begin{array}{l} C_{\tau} \text{ is a } |\mathcal{D}|\text{-dimensional copula,} \\ \forall \tau \in \mathcal{T}(\mathbb{S}). \\ \forall d \in \mathcal{D}, \int_{\mathcal{Y}} \mathbf{g}_{(d)} d\mu(y) = \mathbf{1}_{|\mathcal{T}(\mathbb{S})|}. \\ \exists \pi \in \Pi_{\mathbb{S}} \text{ s.t. } \forall d \in \mathcal{D}, \\ \pi \odot \mathbf{g}_{(d)} \in \Gamma_d(\mathcal{T}(\mathbb{S})). \end{array} \right\}, \end{aligned}$$

where $\mathbf{g}_{(d)(\tau)}$ is the τ -th component of $\mathbf{g}_{(d)}$, $\mathfrak{F}[\mathbf{g}_{(d)(\tau)}]$ is the CDF associated with the density $\mathbf{g}_{(d)(\tau)}$, and $\odot$ denotes the coordinate-wise product of two vectors (Hadamard product).

Proof. Follows from the proof of Theorem 3.3. See Appendix A.3 for details. $\blacksquare$

In other words, treatment selection models do not provide any information about the joint distribution of potential outcomes *within* response-type, but can provide non-trivial information about the joint distribution when aggregated over response-types. Intuitively, this follows from the fact that Condition *M* implies that treatment is randomly assigned conditional on response-type. This also suggests a trade-off: settings that allow for more response-types generate finer partitions of $\operatorname{supp}(\mathbf{V})$ and can, therefore, potentially provide more information on the dependence between outcomes; of course, if there is not enough exogenous variation, any potential gains are offset by the increased degree of underidentification of the marginals and mixing weights (response-type probabilities) themselves.

²²This follows from Proposition 1.2 in Billera and Sturmfels (1992), see the proof of Lemma 3.1 for details.

Remark 3.11. Theorem 3.6 also suggests that the framework here can be used to consider parameters of the form $E(Y_d | Y_{d'} \leq c)$. I do not pursue it formally in this draft, but the basic idea is the same as the extrapolation extension considered in Remark 2.1: Theorem 3.6 can be viewed as saying that, conditional on $\mathbf{T}$, we do not have to worry about any “pointwise” restrictions on the *dependence structure* of $(Y_d, Y_{d'}) | \mathbf{T}$; in other words, the only restrictions on the joint distribution $(Y_d, Y_{d'}) | \mathbf{T}$ are the restrictions on its marginals, $Y_d | \mathbf{T}$ and $Y_{d'} | \mathbf{T}$. So, for the purposes of the Y_d distribution, we can define types based on $(\mathbf{T}, \mathbb{1}\{Y_{d'} < c\})$, for which the identified set can be found using the methods above (it is a function of the $(Y_{d'}, \mathbf{T})$ distribution), and so, we can use the same technique used here to get the identified set for functionals of $(Y_d, \mathbf{T})$ to get the identified set for $(Y_d, (\mathbf{T}, \mathbb{1}\{Y_{d'} < c\}))$. ♦

4. Inference

In this section I assume that the given treatment selection model of interest, $\mathbb{S}$, satisfies the following condition. In the remainder, I will continue to treat $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ as known.

Condition C. $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ is a convex polytope.

Any selection model that simply eliminates response-types (e.g., Ordered/Unordered/Vector/Partial Monotonicity) automatically satisfies this condition. More generally, Condition C is satisfied by any selection model $\mathbb{S}$ such that the (*a priori*) restrictions on the type distribution imposed by $\mathbb{S}$ can be characterized by a set of linear (in)equalities involving response-type probabilities.²³ The main utility of Condition C is tractable computation of the support function of $\Pi_{\mathbb{S}}$, since $\Pi_{\mathbb{S}}$ is a convex polytope whenever $\mathbb{S}$ satisfies Condition C.

Remark 4.1. I conjecture that (for finite $\mathcal{Z}$ and $\mathcal{D}$), under Condition M, Assumption 2.1 in Lee and Salanié (2018) implies Condition C, but I have not been able to prove this. If true, this would enable their broad class of threshold-crossing-rules-based models to be covered by the inference approach outlined here. ♦

4.1. Sharp Specification Test for Treatment Selection Models. ²⁴

To formalize our inference problem, let $\mathcal{P}$ be the set of all probability distributions on $\mathcal{Y} \times \mathcal{D} \times \mathcal{Z}$ satisfying Condition D, and let

$$\mathcal{P}_{\mathbb{S}} := \{P \in \mathcal{P} : \mathcal{Q}_{\mathbb{S}}(P) \neq \emptyset\}.$$

Our hypothesis testing problem is

$$H_0 : P \in \mathcal{P}_{\mathbb{S}} \quad \text{against} \quad H_1 : P \in \mathcal{P} \setminus \mathcal{P}_{\mathbb{S}} \quad (5)$$

In order to use Theorem 3.3 to test if $P \in \mathcal{P}_{\mathbb{S}}$, we need to check

- (i) For each $d \in \mathcal{D}$, for all $y \in \mathcal{Y}$, $\mathcal{B}_{\mathbf{R}_d, \mathbf{T}(\mathbb{S})}(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$
- (ii) $\bigcap_{d \in \mathcal{D}} \Pi_{\mathbf{T}(\mathbb{S})|d}(P) \cap \mathcal{F}_{\mathbf{T}}(\mathbb{S}) \neq \emptyset$

²³In this case, $\mathcal{F}_{\mathbf{T}}(\mathbb{S})$ is simply the intersection of the standard $\mathcal{D}^{\mathcal{Z}}$ -simplex with the polytope defined by the non-negative solutions to such linear (in)equalities.

²⁴The tests proposed in the next two subsections are preliminary and meant as a proof-of-concept. Several optimizations are possible, but I have not yet pursued them in this draft (see Remark 4.2).

Of course, since $\Pi_{\mathcal{T}|d} = \emptyset$ whenever there is some $y \in \mathcal{Y}$ such that $\mathcal{B}_{\mathbf{R}_d, \mathcal{T}}(\mathbf{p}\mathbf{f}_d(y)) = \emptyset$, (ii) implies (i). The distinction is important for implementation, however, because if (i) holds we can use Lemma 3.1 to construct a test of (ii).

Now, note that $\mathcal{B}_{\mathbf{R}_d, \mathcal{T}(\mathbb{S})}(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ if and only if $\mathbf{p}\mathbf{f}_d(y) \in \text{cone}(\mathbf{R}_d, \mathcal{T}(\mathbb{S}))$, where $\text{cone}(\mathbf{R}_d)$ is the cone generated by non-negative linear combinations of the column vectors of $\mathbf{R}_d$,

$$\text{cone}(\mathbf{R}_d) := \{\mathbf{b} \in \mathbb{R}^Z : \exists \mathbf{x} \in \mathbb{R}_{\geq 0}^{\mathcal{T}}, \mathbf{R}_d \mathbf{x} = \mathbf{b}\}.$$

In some cases, this step can be omitted because it reduces to testing $\mathbf{p}\mathbf{f}_d \geq 0$ (which is automatically satisfied).²⁵ More generally, this is a polyhedral cone inclusion problem that can be represented as conditional moment inequality restrictions, as shown in the next result.

Lemma 4.1. *Let $P \in \mathcal{P}$ be given. For any $\mathcal{T} \subseteq \mathcal{D}^Z$ and $d \in \mathcal{D}$, there exist, for some $K_d \in \mathbb{N}$, known functions $\{c_{z,j}^{(d;\mathcal{T})}(P_Z) \mid z \in \mathcal{Z}, j \in \{1, \dots, K_d\}\}$ such that*

$$\begin{aligned} & \mathcal{B}_{\mathbf{R}_d, \mathcal{T}}(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset \quad \forall y \in \mathcal{Y} \\ \iff & \max_{j \in \{1, \dots, K_d\}} \sup_{y \in \mathcal{Y}} E_P \left[\sum_{z \in \mathcal{Z}} c_{z,j}^{(d;\mathcal{T})}(P_Z) \mathbb{1}\{D=d, Z=z\} \mid Y=y \right] \leq 0. \end{aligned}$$

Proof. See Appendix B.1. ■

The proof of Lemma 4.1 generalizes the approach employed by Mourifié and Wan (2017) to test LATE assumptions in the binary treatment setting. Next, we have the following result for testing (ii), which implies that, conditional on $\mathcal{B}_{\mathbf{R}_d, \mathcal{T}}(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ holding for all $y \in \mathcal{Y}$, (ii) can be tested by evaluating $\int_{\mathcal{Y}} h_{\mathbf{R}_1}(\mathbf{c}; \mathbf{p}\mathbf{f}_1(y)) d\mu(y)$ at finitely many known values of $\mathbf{c}$, for each $d \in \mathcal{D}$. In practice, the matrices $\mathbf{M}_d$ can be found by row-reducing $\mathbf{R}_d$ and eliminating pivot variables.

Lemma 4.2. *Let $P \in \mathcal{P}$ be given. For any treatment selection model $\mathbb{S}$ that satisfies Condition C, there exist known (i.e., known from model, does not depend on P) matrices $\mathbf{M}_d$, for $d \in \mathcal{D}$, known matrix $\mathbf{S}$, and known vector $\mathbf{s}$ such that*

$$\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \cap \mathcal{F}_{\mathcal{T}}(\mathbb{S}) \neq \emptyset \iff \exists \mathbf{x} \in [0, 1]^{|\mathcal{T}(\mathbb{S})|} \text{ such that } \begin{bmatrix} \mathbf{M}_1 \\ \vdots \\ \mathbf{M}_{N_D} \\ \mathbf{S} \end{bmatrix} \mathbf{x} \leq \mathbf{h}_{\mathbb{S}}(P),$$

where

$$\mathbf{h}_{\mathbb{S}}(P) := \begin{bmatrix} \left(\int_{\mathcal{Y}} h_{\mathbf{R}_1}(\mathbf{M}_{1,(i)}; \mathbf{p}\mathbf{f}_1(y)) d\mu(y) : i \in [\text{rows}(\mathbf{M}_1)] \right) \\ \vdots \\ \left(\int_{\mathcal{Y}} h_{\mathbf{R}_{N_D}}(\mathbf{M}_{N_D,(i)}; \mathbf{p}\mathbf{f}_{N_D}(y)) d\mu(y) : i \in [\text{rows}(\mathbf{M}_{N_D})] \right) \\ \mathbf{s} \end{bmatrix},$$

and the support function of an empty set is taken to be constant at $-\infty$.

²⁵For example, if the identity matrix is a sub-matrix of $\mathbf{R}_d, \mathcal{T}(\mathbb{S})$, then $\mathcal{B}_{\mathbf{R}_d, \mathcal{T}(\mathbb{S})}(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$ is automatically satisfied.

Proof. See Appendix B.1. ■

Finally, I propose the following test statistic for specification testing treatment selection models.

Proposition 4.1. *For any treatment selection model $\mathbb{S}$ that satisfies Condition C, there exist known vectors $\mathbf{c}_1, \dots, \mathbf{c}_K$, for some $K \geq 1$, such that the hypothesis testing problem (5) is equivalent to testing*

$$H_0 : \vartheta_{\mathbb{S}} \leq 0 \quad \text{against} \quad H_1 : \vartheta_{\mathbb{S}} > 0$$

where

$$\vartheta_{\mathbb{S}} := \max \left\{ \sup_{(y,d,j) \in \mathcal{Y} \times \mathcal{D} \times [K_d]} E_P \left[\sum_{z \in \mathcal{Z}} c_{z,j}^{(d; \mathcal{T}(\mathbb{S}))} (P_Z) \mathbb{1}\{D=d, Z=z\} \mid Y=y \right], \right. \\ \left. \max_{1 \leq i \leq K} \mathbf{c}_i^{\top} \mathbf{h}_{\mathbb{S}}(P) \right\},$$

where the functions $c_{z,j}^{(d; \mathcal{T}(\mathbb{S}))}$ are as given in Lemma 4.1, and $\mathbf{h}_{\mathbb{S}}$ is as given in Lemma 4.2.

Proof. See Appendix B.1. ■

The main advantage of the specification in Proposition 4.1 is implementation, since they are of the form considered by the intersectional bounds framework of Chernozhukov et al. (2013). To implement such a test in practice, we may need to restrict $\mathcal{P}$ further. For example, if Y is continuous, then we will generally at least require $\mathcal{P}$ to be such that $\mathbf{p}\mathbf{f}_d$ has twice differentiable components and is bounded away from zero. Work to formalize these conditions is in progress.

4.2. Inference on Mean Counterfactual Outcomes and Treatment Effects. Fix some continuous $\kappa : \mathcal{Y} \rightarrow \mathbb{R}$ such that $\kappa(\mathcal{Y})$ is a convex compact set. Define

$$\boldsymbol{\theta}_d := (P(T = \tau) E(\kappa(Y_d) \mid T = \tau) : \tau \in \mathcal{T}(\mathbb{S})) ,$$

and let $\boldsymbol{\theta}_{d,Q} := (Q(T = \tau) E_Q(\kappa(Y_d) \mid T = \tau) : \tau \in \mathcal{T}(\mathbb{S}))$.

In this section, I propose methodology for performing inference on parameters of the form

$$\frac{\sum_{d \in \mathcal{D}} \mathbf{c}_d^{\top} \boldsymbol{\theta}_d}{\mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}} \quad \text{or} \quad \sum_{d \in \mathcal{D}} \mathbf{c}_d^{\top} \boldsymbol{\theta}_d \quad \text{or} \quad \mathbf{c}_{\pi}^{\top} \boldsymbol{\pi} ,$$

for some known $\mathbf{c}_d \in \mathbb{R}^{|\mathcal{T}|}$, for $d \in \mathcal{D}$, and $\mathbf{c}_{\pi} \in \mathbb{R}^{|\mathcal{T}|}$. Such parameters include (weighted/unweighted) treatment effects, outcome means, and response-type probabilities. For simplicity, I focus on the parameter

$$\beta := \frac{\mathbf{c}_d^{\top} \boldsymbol{\theta}_d - \mathbf{c}_{d'}^{\top} \boldsymbol{\theta}_{d'}}{\mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}} ,$$

but the results extend analogously to the more general case. I assume that $\mathbf{c}_{\pi}^{\top} \boldsymbol{\pi} > 0$ for all $\boldsymbol{\pi} \in \Pi_{\mathbb{S}}$.

A simple corollary of Theorem 3.4 is that

$$\Theta_{\mathbb{S}}(\beta) := \left\{ \frac{\mathbf{c}_d^{\top} \boldsymbol{\theta}_{d,Q} - \mathbf{c}_{d'}^{\top} \boldsymbol{\theta}_{d',Q}}{\mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}_Q} \mid Q \in \mathcal{Q}_{\mathbb{S}} \right\} = \left\{ \frac{\mathbf{c}_d^{\top} \boldsymbol{\theta}_d - \mathbf{c}_{d'}^{\top} \boldsymbol{\theta}_{d'}}{\mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}} \mid \begin{array}{l} (\boldsymbol{\theta}_d, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} , \\ (\boldsymbol{\theta}_{d'}, \boldsymbol{\pi}) \in \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \cap \tilde{\Pi}_{\mathbb{S}} \end{array} \right\} .$$

Let $\mathcal{P}_{\mathbb{S}} = \{P \in \mathcal{P} : \mathcal{Q}_{\mathbb{S}}(P) \neq \emptyset\}$, as in the previous subsection, and define $\mathcal{P}_{\mathbb{S}}(\beta; \gamma)$, for each $\gamma \in \mathbb{R}$, as

$$\mathcal{P}_{\mathbb{S}}(\beta; \gamma) := \{P \in \mathcal{P}_{\mathbb{S}} : \gamma \in \Theta_{\mathbb{S}}(\beta)\} .$$

For a given value of $\gamma \in \mathbb{R}$, we are interested in the following hypothesis testing problem

$$H_0 : P \in \mathcal{P}_{\mathbb{S}}(\beta; \gamma) \quad \text{against} \quad H_1 : P \in \mathcal{P}_{\mathbb{S}} \setminus \mathcal{P}_{\mathbb{S}}(\beta; \gamma) \quad (6)$$

To proceed, first note that

$$\Theta_{\mathbb{S}}(\beta) = \left\{ b \in \mathbb{R} \left| \begin{array}{l} \exists (\boldsymbol{\theta}_d, \boldsymbol{\pi}_d, \boldsymbol{\theta}_{d'}, \boldsymbol{\pi}_{d'}) \in \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \times \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa) \text{ s.t. ,} \\ \boldsymbol{\pi}_d = \boldsymbol{\pi}_{d'} \in \Pi_{\mathbb{S}}, \\ \mathbf{c}_d^{\top} \boldsymbol{\theta}_d - \mathbf{c}_{d'}^{\top} \boldsymbol{\theta}_{d'} = b \mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}_d \end{array} \right. \right\} .$$

Next, motivated by this definition, define

$$\Lambda_{\mathbb{S}}(\beta; \gamma) := \left\{ (\boldsymbol{\theta}_d, \boldsymbol{\pi}_d, \boldsymbol{\theta}_{d'}, \boldsymbol{\pi}_{d'}) \left| \begin{array}{l} \boldsymbol{\theta}_d, \boldsymbol{\theta}_{d'} \in \kappa(\mathcal{Y})^{|\mathcal{T}|}, \boldsymbol{\pi}_d, \boldsymbol{\pi}_{d'} \in \Pi_{\mathbb{S}} \\ \boldsymbol{\pi}_d = \boldsymbol{\pi}_{d'}, \mathbf{c}_d^{\top} \boldsymbol{\theta}_d - \mathbf{c}_{d'}^{\top} \boldsymbol{\theta}_{d'} = \gamma \mathbf{c}_{\pi}^{\top} \boldsymbol{\pi}_d \end{array} \right. \right\} .$$

Note that when $\mathbb{S}$ satisfies Condition C, $\Lambda_{\mathbb{S}}(\beta; \gamma)$ is a compact convex polytope. Denote by $h_{\Lambda_{\mathbb{S}}(\beta; \gamma)}$ the support function of $\Lambda_{\mathbb{S}}(\beta; \gamma)$. Now, we have the following result.

Proposition 4.2. *For any treatment selection model $\mathbb{S}$ that satisfies Condition C, the hypothesis testing problem (6) is equivalent to*

$$H_0 : \sup_{\mathbf{v} \in S^{4|\mathcal{T}(\mathbb{S})|}} \vartheta_{(\beta; \gamma)}(\mathbf{v}) \leq 0 \quad \text{against} \quad H_1 : \sup_{\mathbf{v} \in S^{4|\mathcal{T}(\mathbb{S})|-1}} \vartheta_{(\beta; \gamma)}(\mathbf{v}) > 0 ,$$

where $S^{4|\mathcal{T}(\mathbb{S})|-1}$ is the unit sphere in $\mathbb{R}^{4|\mathcal{T}(\mathbb{S})|}$, and

$$\begin{aligned} \vartheta_{(\beta; \gamma)}(\mathbf{v}) := & - \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\kappa(y) \mathbf{v}_1 + \mathbf{v}_2 ; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) \\ & - \int_{\mathcal{Y}} h_{\mathbf{R}_{d'}}(\kappa(y) \mathbf{v}_3 + \mathbf{v}_4 ; \mathbf{p}\mathbf{f}_{d'}(y)) d\mu(y) \\ & - h_{\Lambda_{\mathbb{S}}(\beta; \gamma)}(-\mathbf{v}) , \end{aligned}$$

for all $\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) \in S^{4|\mathcal{T}(\mathbb{S})|}$ with $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^{|\mathcal{T}(\mathbb{S})|}$.

Proof. See Appendix B.2. ■

The test proposed in Proposition 4.2 can be inverted to produce confidence regions for β . Note that it is once again of the intersectional bounds form considered by Chernozhukov et al. (2013) (CLR) and, under some regularity conditions, this test can be implemented using their framework. Note also that if we replace the test statistic in Proposition 4.2 with the maximum of the test statistics in Propositions 4.1 and 4.2, then the joint test will always fail to reject whenever the specification test fails to reject, and so, will return an empty confidence region whenever we can reject $P \in \mathcal{P}_{\mathbb{S}}$.

Remark 4.2. Implementing the test proposed in Proposition 4.2 using CLR's framework may be very computationally demanding when there are many response-types because their approach will typically involve optimization over a grid on $S^{4|\mathcal{T}(\mathbb{S})|-1}$. But $\vartheta_{(\beta; \gamma)}(\mathbf{v})$ is a concave function, and so, computation of the test statistic $\sup_{\mathbf{v} \in S^{4|\mathcal{T}(\mathbb{S})|}} \vartheta_{(\beta; \gamma)}(\mathbf{v})$ is a convex programming problem. In our setting, however, implementation using CLR's methodology

generally cannot take advantage of this fact. This is because their approach rejects H_0 for large values of the following *precision-corrected* estimate of the test statistic,

$$\hat{\vartheta}_0(p) := \sup_{\mathbf{v} \in S^{4|\mathcal{T}(\mathbb{S})|}} \left(\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v}) - k(p) s(\mathbf{v}) \right),$$

where $\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v})$ is an estimator for $\vartheta_{(\beta;\gamma)}(\mathbf{v})$, $k(p)$ is a critical value, and $s(\mathbf{v})$ is a (point-wise) standard error for $\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v})$. If $\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v})$ is a plug-in estimator for $\vartheta_{(\beta;\gamma)}(\mathbf{v})$ based on, e.g., a kernel density estimator, then $\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v})$ remains a concave function, but the addition of the adjustment term $-k(p) s(\mathbf{v})$ generally renders $\hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v}) - k(p) s(\mathbf{v})$ non-concave.

The precision-correction in CLR is due to the fact that, in the problems that motivate their work, the analog estimator $\sup_{\mathbf{v} \in S^{4|\mathcal{T}(\mathbb{S})|}} \hat{\vartheta}_{(\beta;\gamma)}(\mathbf{v})$ can be severely biased upward in finite samples. Informally, this is because in settings like testing conditional moment inequalities (e.g.), the bound function $\vartheta_{(\beta;\gamma)}(\mathbf{v})$ has a “local character” (this is also true, for example, for the test statistic proposed in Proposition 4.1). On the other hand, this may not be a concern in our setting since the estimation of each of the three components of $\vartheta_{(\beta;\gamma)}(\mathbf{v})$ is perhaps best understood as the estimation of a finite dimensional polytope, which we should not expect to have such problems. Indeed, it is possible to show that the estimator, $\hat{h}_{\mathbf{R}_d}$, for $h_{\mathbf{R}_d}$ constructed by plugging in a suitably undersmoothed (and sufficiently high order) kernel density estimator, $\mathbf{p}\mathbf{f}_d$, for $\mathbf{p}\mathbf{f}_d$, consistently (and uniformly) estimates the function $h_{\mathbf{R}_d}$, and $\sqrt{n}(\hat{h}_{\mathbf{R}_d} - h_{\mathbf{R}_d})$ converges in law, as a stochastic process, to a (non-degenerate) asymptotic distribution. This suggests that strong approximation methods are not necessary to implement a test based on Proposition 4.2.

To take advantage of these properties, we can use an alternate implementation based on the Delta method for directionally differentiable functionals, as provided in Fang and Santos (2019), and the numerical derivative approach developed in Hong and Li (2018), using the kernel CDF²⁶ as the underlying basis for the functional central limit theorem (as in, e.g., Aït-Sahalia, 1993). As suggested by Hong and Li (2018), inference based on this approach will, under some regularity conditions, be uniformly valid over a large class of data distributions. $\blacklozenge$

5. Implementation

To implement the identification results obtained above in practice, we need a computationally tractable approach to compute the support functions given by Lemmas 3.1 and 3.2:

$$h_{\Pi_{\mathcal{T}_d}}(\mathbf{c}) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y), \quad (7)$$

$$\text{and } h_{\Theta_{\mathcal{T}_d}(\kappa)}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\kappa(y) \mathbf{c}_1 + \mathbf{c}_2; \mathbf{p}\mathbf{f}_d(y)) d\mu(y). \quad (8)$$

The expressions above involve the function $(\mathbf{c}, y) \mapsto h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y))$. For fixed $\mathbf{c} \in \mathbb{R}^{|\mathcal{T}|}$ and $y \in \mathcal{Y}$, $h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y))$ is a “small” linear programming problem. Keeping $\mathbf{c}$ fixed, the function $y \mapsto h_{\mathbf{R}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y))$ is the solution to a *parametric* linear programming problem (parametric in the right hand side vector). When there are only a handful of distinct instrument levels, the problem can often be solved symbolically²⁷ to obtain analytic solutions. Such

²⁶Defined as $y \mapsto \int_{[\min \mathcal{Y}, y]} \mathbf{p}\hat{\mathbf{f}}_d(u) d\mu(u)$.

²⁷This can be done either “manually” using, e.g., Fourier-Motzkin elimination, or using symbolic solvers like Mathematica or Maple

analytic solutions are useful for developing intuition on the source of identification power, for performing inference, and for implementing sophisticated forms of covariate adjustment.

As in many settings, the analytic approach becomes increasingly complicated as the parameter space grows richer²⁸. Indeed, the complexity of the analytic solution (and the difficulty of deriving it) generally increases exponentially with the cardinality of $\mathcal{Z}$, and it can be computationally prohibitive to solve symbolically when there are many instruments/response-types²⁹. In such cases, $h_{\mathbf{R}_d}$ can still be easily approximated using a primal-dual approach (with deterministic bounds on the approximation error); in this section, I provide a general algorithm to implement such an approach for computing $h_{\mathbf{R}_d}$. Importantly, the proposed algorithm can provide an *exact solution after finitely many steps*³⁰. When there are many instruments, however, the number of steps may be too high for practical purposes, and we may wish to rely on approximations of $h_{\mathbf{R}_d}$.

5.1. Computation of $h_{\mathbf{R}_d}$.

Notation and Preliminaries: Basic Solutions to Linear Programs. For some $\mathbf{b} \in \mathbb{R}^m$, $\mathbf{c} \in \mathbb{R}^n$, and $\mathbf{A} \in \mathbb{R}^{m \times n}$, with $\text{rank}(\mathbf{A}) = m$, consider the following linear program (LP):

$$\max \mathbf{c}^\top \mathbf{x} \quad \text{subject to} \quad \mathbf{A}\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}$$

For each subset $B \subseteq \{1, \dots, n\}$ of columns of $\mathbf{A}$, let $\mathbf{A}_B$ be the sub-matrix formed by the columns in B . If $\mathbf{A}_B^{-1}$ is invertible, we will say that B is a basis. For any basis $B = \{j_1, \dots, j_m\}$, with $j_1 < j_2 < \dots < j_m$, such that $\mathbf{A}_B^{-1} \mathbf{b} \geq \mathbf{0}$, we can construct a feasible point $\mathbf{x}$, as follows: set $x_i = 0$ for all $i \notin B$, and set each x_{j_k} to the k -th value of $\mathbf{A}_B^{-1} \mathbf{b}$. Such a feasible point is called a *basic feasible solution*. Suppose the LP has at least one solution, and that $\mathbf{c}^\top \mathbf{x}$ is bounded over the feasible space. Then, there always exists a basis $B^*(\mathbf{c}; \mathbf{b})$ such that the basic feasible solution associated with the basis $B^*(\mathbf{c}; \mathbf{b})$ is an *optimal* solution to the above LP. This is a well-known result in linear programming (see, e.g., Theorem 4.2.3 in Matousek and Gärtner, 2007). This result also continues to hold if the LP above is a minimization problem instead of a maximization problem.

The LP above (called the *primal* LP) has an associated *dual* LP:

$$\min \mathbf{b}^\top \mathbf{y} \quad \text{subject to} \quad \mathbf{A}^\top \mathbf{y} - \mathbf{s} = \mathbf{c}, \mathbf{s} \geq \mathbf{0},$$

²⁸For example, see the brief discussion in Torgovitsky (2019a)

²⁹Analytic solutions may still remain feasible as the number of instrument-levels increase if we also impose increasingly severe restrictions on response-types, so that the “degree of underidentification” remains controlled.

³⁰Both the symbolic and computational approaches are tackling a finite problem: polytopes of the form $K_{\mathbf{b}} := \{\mathbf{x} \geq \mathbf{0} : \mathbf{R}_d \mathbf{x} = \mathbf{b}\}$ will take on one of finitely many “shapes” as the right hand side vector $\mathbf{b}$ is allowed to vary. More precisely, every polytope $K_{\mathbf{b}}$ generates a (finite) decomposition (called the *normal fan* of $K_{\mathbf{b}}$) of the space of linear functionals on $K_{\mathbf{b}}$ into functionals that are maximized on the same face of $K_{\mathbf{b}}$. If $K_{\mathbf{b}}$ and $K_{\mathbf{b}'}$ generate the same decomposition, they are said to be *normally equivalent* (they have the same “shape”); when this is the case, we can view maximizing $\mathbf{c}^\top \mathbf{x}$ over $K_{\mathbf{b}}$ and over $K_{\mathbf{b}'}$ as the “same” problem (in a formal sense). The “shape” of $K_{\mathbf{b}}$ corresponds to which inequalities are non-redundant in the halfspace representation of $K_{\mathbf{b}}$, and so, the number of possible shapes generally grows with $\mathcal{Z}$. For more details, I refer the interested reader to Section 1.2 in De Loera, Rambau, and Santos (2010), and references therein.

Since each decomposition is finite and since, for any fixed matrix $\mathbf{R}_d$, the space of right hand side vectors only generate finitely many “shapes”, the problem of computing $h_{\mathbf{R}_d}(\cdot; \cdot)$ is finite (at least when $\mathbf{p}\mathbf{f}_d$ is continuous and $\mathcal{Y}$ compact). Although both approaches deal with finite problems, the symbolic solution amounts to solving the problem for all possible observed data distributions, whereas the computational approach only needs to solve it for the actually observed data distribution (which may generate far fewer “shapes” than the maximum possible).

which can equivalently be represented in the same form as the original LP by writing:

$$\min \begin{bmatrix} \mathbf{b} \\ -\mathbf{b} \\ \mathbf{0} \end{bmatrix}^\top \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{s} \end{bmatrix} \quad \text{subject to} \quad [\mathbf{A}^\top - \mathbf{A}^\top - \mathbf{I}_n] \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{s} \end{bmatrix} = \mathbf{c}, \quad \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \mathbf{s} \end{bmatrix} \geq \mathbf{0},$$

and so, the notion of optimal basis also makes sense for the dual LP. The weak duality theorem, a well-known result in linear programming, states that for every feasible solution, $\mathbf{x}$, to the primal LP, and every feasible solution, $\mathbf{y}$, to the dual LP, we have that $\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}$. Further, the strong duality theorem states that at optimal solutions, $\mathbf{x}^*$ and $\mathbf{y}^*$, to the primal and dual (respectively), we have that $\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*$.

In our case, we are interested in LPs of the form

$$\max \kappa(y)^\top \mathbf{x} \quad \text{subject to} \quad \mathbf{R}_d \mathbf{x} = \mathbf{p}\mathbf{f}_d(y), \quad \mathbf{x} \geq \mathbf{0},$$

where $\kappa : \mathcal{Y} \rightarrow \mathbb{R}^{|\mathcal{T}|}$ is a known function, generally of the form $\kappa(y) = \mathbf{c}$ or $\kappa = \kappa(y) \mathbf{c}_1 + \mathbf{c}_2$. Denote by $B_{\kappa}^{\text{prim}}(y)$ the optimal basis associated with this problem, and denote by $B_{\kappa}^{\text{dual}}(y)$ the optimal basis associated with the dual of this problem; let $\mathbf{S}_{d, B_{\kappa}^{\text{prim}}(y)}^{\text{prim}}$ and $\mathbf{S}_{d, B_{\kappa}^{\text{dual}}(y)}^{\text{dual}}$ be the matrices such that the primal and dual solutions are given by $\mathbf{S}_{d, B_{\kappa}^{\text{prim}}(y)}^{\text{prim}} \mathbf{p}\mathbf{f}_d(y)$ and $\mathbf{S}_{d, B_{\kappa}^{\text{dual}}(y)}^{\text{dual}} \kappa(y)$, respectively. The primal and dual optimal bases can be found simultaneously via linear programming;³¹ for any fixed y , the corresponding LP is very small in the settings that motivate this paper.

5.1.1. Generic Computation Algorithm. For simplicity, I focus on the case where Y is a scalar and $\mathcal{Y} \subseteq \mathbb{R}$ is a compact interval.

Remark 5.1. The algorithm proposed here is preliminary and meant as a proof-of-concept. Many optimizations are possible (for example, using a parametric LP approach), but I have not yet pursued them in this draft. The algorithm also abstracts away from how the actual linear programming step is performed, but certain kinds of “pre-processing” (e.g., eliminating redundant response-types etc.), can often significantly reduce the size of the problem. ♦

Algorithm 5.1.

Input: Model and Data: binary matrix $\mathbf{R}_d$ and continuous function $\mathbf{p}\mathbf{f}_d : \mathcal{Y} \rightarrow \text{cone}(\mathbf{R}_d)$.

Parameters: continuous function $\kappa : \mathcal{Y} \rightarrow \mathbb{R}^{|\mathcal{T}|}$, real number $\delta \geq 0$, and finite set of grid points $\mathcal{G} \subseteq \mathcal{Y}$ with $\{\min \mathcal{Y}, \max \mathcal{Y}\} \in \mathcal{G}$.

Output: $(\mathbf{h}^{\text{prim}}, \mathbf{h}^{\text{dual}})$, two measurable functions in $\{\mathcal{Y} \rightarrow \mathbb{R}\}$.

Step 0. Set $\mathcal{G}_1 := \mathcal{G}$.

$\vdots$

Step j . For each $y \in \mathcal{G}_j$, find $B_{\kappa}^{\text{prim}}(y)$ and $B_{\kappa}^{\text{dual}}(y)$ ³².

³¹Throughout this section, I assume that the LPs of interest are bounded, and therefore have solutions; while this is not necessarily true (the LP will be unbounded iff the objective function assigns a positive coefficient to a response-type that never takes treatment d), the degenerate situations that involve such problems are easy to handle separately.

³²This is the linear programming step, and can be performed with any linear program solver that outputs the optimal bases (in addition to the optimal solution), e.g., Gurobi.

For each $y \in \mathcal{Y}$ define $\lfloor y \rfloor, \lceil y \rceil \in \mathcal{G}_j$ as the tightest values such that $\lfloor y \rfloor \leq y \leq \lceil y \rceil$, and define $\mathbf{s}_y := \max_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\}} \left\{ \min \left(\mathbf{S}_{d, B_{\mathbf{x}}^{\text{prim}}(\tilde{y})}^{\text{prim}} \mathbf{p}\mathbf{f}_d(y) \right) \right\}$.

(Feasibility Check) If $\inf_{y \in \mathcal{Y}} \mathbf{s}_y < 0$:

There exists $y_0 \leq \dots \leq y_k \in \mathcal{Y}$, for some $k \geq 1$, such that

$$\{y \in \mathcal{Y} \mid \mathbf{s}_y < 0\} = \bigcup_{n=1}^k (y_{n-1}, y_n) .$$

Set $\mathcal{G}_{j+1} := \mathcal{G}_j \cup \left\{ \frac{y_n - y_{n-1}}{2} : n \in [k] \right\}$, and immediately go to the next step.

Else, if $\inf_{y \in \mathcal{Y}} \mathbf{s}_y \geq 0$:

Order elements of $\mathcal{G}_j$ as $\inf \mathcal{Y} =: y_0 \leq \dots \leq y_N := \sup \mathcal{Y}$ and compute

$$\begin{aligned} \mathbf{h}_j^{\text{prim}}(y) &:= \sum_{n=1}^N \mathbb{1}_{\{y \in [y_{n-1}, y_n]\}} \max_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\} : \mathbf{s}_{\tilde{y}} \geq 0} \left\{ \varkappa(y)^\top \mathbf{S}_{d, B_{\mathbf{x}}^{\text{prim}}(\tilde{y})}^{\text{prim}} \mathbf{p}\mathbf{f}_d(y) \right\} , \\ \mathbf{h}_j^{\text{dual}}(y) &:= \sum_{n=1}^N \mathbb{1}_{\{y \in [y_{n-1}, y_n]\}} \min_{\tilde{y} \in \{\lfloor y \rfloor, \lceil y \rceil\} : \mathbf{s}_{\tilde{y}} \geq 0} \left\{ \mathbf{p}\mathbf{f}_d(y)^\top \mathbf{S}_{d, B_{\mathbf{x}}^{\text{dual}}(\tilde{y})}^{\text{dual}} \varkappa(y) \right\} . \end{aligned}$$

(Optimality Check) If $\sup_{y \in \mathcal{Y}} \left(\mathbf{h}_j^{\text{dual}}(y) - \mathbf{h}_j^{\text{prim}}(y) \right) > \delta$:

There exists $y_0 \leq \dots \leq y_k$, for some $k \geq 1$, such that

$$\left\{ y \in \mathcal{Y} \mid \left(\mathbf{h}_j^{\text{dual}}(y) - \mathbf{h}_j^{\text{prim}}(y) \right) > \delta \right\} = \bigcup_{n=1}^k (y_{n-1}, y_n) .$$

Set $\mathcal{G}_{j+1} := \mathcal{G}_j \cup \left\{ \frac{y_n - y_{n-1}}{2} : n \in [k] \right\}$, and immediately go to the next step.

Else, if $\sup_{y \in \mathcal{Y}} \left(\mathbf{h}_j^{\text{dual}}(y) - \mathbf{h}_j^{\text{prim}}(y) \right) \leq \delta$: Output $(\mathbf{h}_j^{\text{prim}}, \mathbf{h}_j^{\text{dual}})$ and stop.

⋮

Now, we have the following result.

Proposition 5.1. Assume that $\mathcal{Y} \subseteq \mathbb{R}$ is a compact interval, that μ is the Lebesgue measure, and that $\mathbf{p}\mathbf{f}_d$ is continuous. For any treatment selection model $\mathbb{S}$ such that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$, and any continuous function $\varkappa : \mathcal{Y} \rightarrow \mathbb{R}^{|\mathcal{T}|}$, real number $\delta \geq 0$, and finite set of grid points $\mathcal{G} \subseteq \mathcal{Y}$ such that $\{\min \mathcal{Y}, \max \mathcal{Y}\} \in \mathcal{G}$, the following statements hold for Algorithm 5.1:

- (i) Algorithm 5.1 will terminate in finitely many steps, and output two measurable functions $(\mathbf{h}^{\text{prim}}, \mathbf{h}^{\text{dual}})$ in $\{\mathcal{Y} \rightarrow \mathbb{R}\}$ such that

$$\sup_{y \in \mathcal{Y}} \left(\mathbf{h}^{\text{dual}}(y) - \mathbf{h}^{\text{prim}}(y) \right) \leq \delta$$

(ii) Algorithm 5.1's output $(\mathbf{h}^{prim}, \mathbf{h}^{dual})$ will always be such that, for all $y \in \mathcal{Y}$

$$\mathbf{h}^{prim}(y) \leq h_{\mathbf{R}_d}(\boldsymbol{\varkappa}(y); \mathbf{p}\mathbf{f}_d(y)) \leq \mathbf{h}^{dual}(y)$$

Further, for any such selection model, for every $(\mathbf{c}_1, \mathbf{c}_2) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}^{|\mathcal{T}|}$ and every continuous $\kappa : \mathcal{Y} \rightarrow \mathbb{R}$, there exists a finite set $\mathcal{G}^* \subseteq \mathcal{Y}$ such that with input parameters $\boldsymbol{\varkappa}(y) = \kappa(y) \mathbf{c}_1 + \mathbf{c}_2$, $\delta = 0$, and $\mathcal{G} = \mathcal{G}^*$, Algorithm 5.1 stops after Step 1.

Proof. See Appendix C.1. ■

Remark 5.2. In practice, we do not know $\mathbf{p}\mathbf{f}_d$, and so, we will need to implement Algorithm 5.1 using an estimator, $\hat{\mathbf{p}\mathbf{f}}_d$, for $\mathbf{p}\mathbf{f}_d$ (e.g. kernel density estimator). It may sometimes be the case that $\hat{\mathbf{p}\mathbf{f}}_d \notin \text{cone}(\mathbf{R}_d)$ even though $\mathbf{p}\mathbf{f}_d \in \text{cone}(\mathbf{R}_d)$, which may lead to issues with implementation. In such cases, Algorithm 5.1 can be implemented after projecting $\hat{\mathbf{p}\mathbf{f}}_d$ onto $\text{cone}(\mathbf{R}_d)$. Note that although the specification test proposed in Proposition 4.1 requires the computation of quantities like (7), it is not required for the test proposed in Lemma 4.1 for testing $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset \forall y \in \mathcal{Y}$ (or, equivalently, $\mathbf{p}\mathbf{f}_d \in \text{cone}(\mathbf{R}_d)$), which is all that is necessary for the efficacy of Algorithm 5.1. ◆

6. Conclusion

This paper presents new specification testing and partial identification results for instrumental variable models with multiple levels (or types) of treatment and discrete-valued, randomly assigned, instruments. The proposed framework allows for the partial identification of a broad class of causal parameters under many different kinds of assumptions about the treatment selection mechanism. Other than random assignment of the instrument, I do not require any additional restrictions on treatment selection, but allow the analyst to flexibly model selection into treatment. Using a novel geometric characterization of IV mixture models, this paper provides a characterization of the empirical content and testable implications of such models, and a new characterization of the identified set for counterfactual choices, outcome means, and treatment effects under such models. I propose methods for (i) (sharply) testing treatment selection model specifications and (ii) performing inference on mean causal effects and counterfactual outcomes under such assumptions. Finally, I propose a computationally feasible algorithm for implementing the proposed method without relying on any parametric or semiparametric assumptions.

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Appendix A: Proofs of Main Identification Results

A.1. Proof of Theorem 3.1

Theorem 3.1. *For any treatment selection model $\mathbb{S}$,*

$$\mathcal{Q}_{\mathbb{S}} = \{Q \in \mathcal{Q} \mid \pi_Q \in \mathcal{F}_T(\mathbb{S})\} .$$

Proof. The “ $\subseteq$ ” direction is clear, and so, it suffices to show “ $\supseteq$ ”.

Fix $Q \in \{Q \in \mathcal{Q} : Q_T \in \mathcal{F}_T(\mathbb{G}, \mathcal{V}, \mathcal{F}_V)\}$ and define $\mathcal{T}_Q := \{\tau \in \mathcal{D}^Z : Q(T = \tau) > 0\}$. We have that there exists $\tilde{Q}$ such that $\tilde{Q}_T = Q_T$ and $\tilde{Q}$ satisfies $(\mathbb{G}, \mathcal{V}, \mathcal{F}_V)$. So, there exists a measurable function $g_D \in \mathbb{G}$ and random variable V with $\text{supp}(V) = \mathcal{V}$ and $\text{Law}(V) \in \mathcal{F}_V$ such that $\text{Law}((g_D(z, V) : z \in \mathcal{Z})) = \tilde{Q}_T = Q_T$. Let $Z \sim Q_Z$ with $Z \perp V$ mutually independent.

Now it suffices to show that there exists measurable function g_Y and random variable ε_Y such that Z, V, ε_Y are mutually independent, and, for each $\tau \in \mathcal{T}_Q$,

$$\text{Law}((g_Y(d, V, \varepsilon_Y) : d \in \mathcal{D}) \mid T = \tau) = Q_{(Y_d : d \in \mathcal{D}) \mid T = \tau} .$$

For each $\tau \in \mathcal{T}_Q$, define $\varepsilon_Y(\tau)$ as a $|\mathcal{D}|$ -dimensional random vector with distribution $Q_{(Y_d : d \in \mathcal{D}) \mid T = \tau}$, with $\varepsilon_Y := (\varepsilon_Y(\tau)^\top : \tau \in \mathcal{T}_Q)^\top$ independent from (Z, V) . Next, define $g_Y : \mathcal{D} \times \text{supp}(V) \times \text{supp}(\varepsilon_Y)$

$$g_Y(d, v, e) := e_{(T(v))(d)} ,$$

where $e_{(\tau)(d)}$ is the d -th component of the τ -th tuple of the stacked tuple e .

Finally, define $D = g_D(Z, V)$ and $Y = g_Y(D, V, \varepsilon_Y)$.

By construction, we have that $(Y, D, Z, V, \varepsilon_Y)$ satisfies $(\mathbb{G}, \mathcal{V}, \mathcal{F}_V)$ with the (g_D, g_Y) defined above, and $\text{Law}((g_Y(d, V, \varepsilon_Y) : d \in \mathcal{D}), (g_D(z, V) : z \in \mathcal{Z}), Z) = Q$, implying that Q satisfies $(\mathbb{G}, \mathcal{V}, \mathcal{F}_V)$, as required. ■

A.2. Proof of Theorem 3.2

First, note that the random assignment of Z implies that parent and observed data distributions are related in the following convenient way.

Lemma 1.1. *For any $Q \in \mathcal{M}$, P_Q depends only on Q_Z , $\{Q_{D_z} : z = \mathcal{Z}\}$, and $\{Q_{Y_d \mid D_z = d} : d \in \mathcal{D}, z \in \mathcal{Z}\}$.*

Proof. Let (y, d, z) be an arbitrary element of the support of (Y, D, Z) . Now, we have that, for any $Q \in \mathcal{M}$,

$$\begin{aligned} P_Q(Y \leq y, D = d, Z = z) &= Q(Y_d \leq y, D_z = d) Q(Z = z) \\ &= Q(Y_d \leq y \mid D_z = d) Q(D_z = d) Q(Z = z) , \end{aligned}$$

which immediately implies the result. $\blacksquare$

Theorem 3.2 (Sharp Characterization of $\mathcal{Q}_{\mathcal{T}}$). *Suppose P satisfies Condition D. Then, for any $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$, $Q \in \mathcal{Q}_{\mathcal{T}}$ if and only if $Q \in \mathcal{M}$, $Q_Z = P_Z$ and, for each $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, the following equality holds*

$$\mathbf{R}_d \pi \xi_{Y_d}(y) = \mathbf{p} \mathbf{f}_d(y) . \quad (\star)$$

Proof. ($\Leftarrow$) For any $z \in \mathcal{Z}, d \in \mathcal{D}$ and $y \in \mathcal{Y}$, we have that

$$\begin{aligned} &Q(Y_d \leq y, D_z = d, Z = z) \\ &= Q(Z = z) \int_{u \in \mathcal{Y}: u \leq y} \sum_{\tau \in \mathcal{T}: \tau(z) = d} Q(\mathbf{T} = \tau) f_{d,Q}(u \mid \mathbf{T} = \tau) d\mu(u) \\ &= P(Z = z) P(D = d \mid Z = z) P(Y \leq y \mid D = d, Z = z) \\ &= P(Y \leq y, D = d, Z = z) , \end{aligned}$$

where the second equality follows from integrating both sides of the corresponding equation in the system $(\star)$, multiplying both sides by $P(Z = z)$, and using $Q(Z = z) = P(Z = z)$. This implies that $P_Q = P$, so $Q \in \mathcal{Q}$.

We also have that

$$\begin{aligned} &\sum_{\tau \in \mathcal{T}} Q(\mathbf{T} = \tau) \\ &= \sum_{d \in \mathcal{D}} \sum_{\tau \in \mathcal{T}: \tau(z) = d} Q(\mathbf{T} = \tau) \\ &= \sum_{d \in \mathcal{D}} \int_{y \in \mathcal{Y}} \sum_{\tau \in \mathcal{T}: \tau(z) = d} Q(\mathbf{T} = \tau) f_{d,Q}(y \mid \mathbf{T} = \tau) d\mu(y) \\ &= \sum_{d \in \mathcal{D}} P(D = d \mid Z = z) \\ &= 1 , \end{aligned}$$

so $Q \in \mathcal{Q}_{\mathcal{T}}$, as required.

($\Rightarrow$) For any $Q \in \mathcal{M}$, we have that

$$\sum_{\tau \in \mathcal{D}^{\mathcal{Z}}: \tau(z) = d} Q(\mathbf{T} = \tau) f_{d,Q}(y \mid \mathbf{T} = \tau) = P_Q(D = d \mid Z = z) f_{P_Q}(y \mid D = d, Z = z) .$$

Recall that $Q \in \mathcal{Q}_{\mathcal{T}}$ implies that $P_Q = P$ and $\sum_{\tau \in \mathcal{T}} Q(D_z = \tau(z), \forall z \in \mathcal{Z}) = 1$ for all $z \in \mathcal{Z}$. Since $\mathcal{Q}_{\mathcal{T}} \subseteq \mathcal{M}$, we have that

$$\sum_{\tau \in \mathcal{T}: \tau(z) = d} Q(\mathbf{T} = \tau) f_{d,Q}(y \mid \mathbf{T} = \tau) = P(D = d \mid Z = z) f_P(y \mid D = d, Z = z) ,$$

which immediately implies the result. $\blacksquare$

A.3. Proof of Theorem 3.3

Theorem 3.3. Suppose P satisfies Condition D. Then, for any $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$,

$$\Pi_{\mathcal{T}} = \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d},$$

and $\Pi_{\mathcal{T}}$ is a compact convex polytope in $\mathbb{R}_{\geq 0}^{|\mathcal{T}|}$. Moreover, $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$ if and only if $\Pi_{\mathcal{T}} \neq \emptyset$.

Proof. First, assume that $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$, which implies that $\Pi_{\mathcal{T}} \neq \emptyset$, and fix a vector $p \in \Pi_{\mathcal{T}}$. Now, we must have that there exists $Q \in \mathcal{Q}_{\mathcal{T}}$ such that $\pi_Q = p$, and

$$\sum_{\tau \in \mathcal{T}: \tau(z)=d} \pi_Q(\tau) f_{d,Q}(y | T = \tau) = p_{(d|z)} f_{(d,z)}(y) \quad \forall (d, y, z) \in \mathcal{D} \times \mathcal{Y} \times \mathcal{Z}.$$

For any $d \in \mathcal{D}$, define $g_{(\tau)}^{(d)}(y) := \pi_Q(\tau) f_{d,Q}(y | T = \tau)$ for each $y \in \mathcal{Y}$, and note that $g^{(d)} \in \Gamma(\mathcal{B}_d)$ and $\int_{\mathcal{Y}} g^{(d)}(y) d\mu(y) = \pi_Q = p$, and so, $p \in \Pi_{\mathcal{T}|d}$. This works for any d , and so, $\Pi_{\mathcal{T}} \subseteq \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d}$. Note that we now also have that

$$\mathcal{Q}_{\mathcal{T}} \neq \emptyset \implies \Pi_{\mathcal{T}} \neq \emptyset \implies \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} \neq \emptyset.$$

Next, assume that $\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d} \neq \emptyset$, fix a $p \in \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d}$. Now, we have that, for each $d \in \mathcal{D}$, there exists $g^{(d)} \in \Gamma(\mathcal{B}_d)$ such that $\int_{\mathcal{Y}} g^{(d)}(y) d\mu(y) = p$. Define, for each $d \in \mathcal{D}$ and $\tau \in \mathcal{T}$,

$$f_{\tau}^{(d)}(y) := \frac{g_{(\tau)}^{(d)}(y)}{p(\tau)} = \frac{g_{(\tau)}^{(d)}(y)}{\int_{\mathcal{Y}} g_{(\tau)}^{(d)}(y) d\mu(y)}.$$

Now, $f_{\tau}^{(d)}(y)$ is a valid density function for each $\tau \in \mathcal{T}$ and each $d \in \mathcal{D}$; let $F_{\tau}^{(d)}(y)$ be the CDF associated with this density.

For each $\tau \in \mathcal{D}^{\mathcal{Z}}$, let C_{τ} be some $|\mathcal{D}|$ -dimensional copula and, when $\tau \notin \mathcal{T}$, define $p(\tau) := 0$. Now, define Q as

$$Q(y, \tau, z) := C_{\tau} \left(\left(F_{\tau}^{(d)}(y_{(d)}) : d \in \mathcal{D} \right) \right) \times p(\tau) \times Q_Z(z),$$

for any $y \in \prod_{d \in \mathcal{D}} \mathcal{Y}$, $\tau \in \mathcal{D}^{\mathcal{Z}}$, and $z \in \mathcal{Z}$, where $Q(y, \tau, z)$ is shorthand for

$$Q((Y_d : d \in \mathcal{D}) \leq y, (D_z : z \in \mathcal{Z}) = \tau, Z = z).$$

This construction ensures that $Q \in \mathcal{M}$ and, by Lemma 1.1, that $P_Q = P$. Since $\pi_Q = p$ by construction, and p only has support in $\mathcal{T}$, we have that $Q \in \mathcal{Q}_{\mathcal{T}}$, which implies that $\Pi_{\mathcal{T}} \supseteq \bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}|d}$ and $\mathcal{Q}_{\mathcal{T}} \neq \emptyset$, completing the proof. $\blacksquare$

A.4. Proofs of Lemmas 3.1 and 3.2

A.4.1. Proof of Lemma 3.1.

Proof. First consider the map $y \mapsto \tilde{\mathcal{B}}_d(y)$ where $\tilde{\mathcal{B}}_d(y) := \left\{ x \in \mathbb{R}^{|\mathcal{T}_d|} \mid \tilde{\mathbf{R}}_d x = \mathbf{p} f_d(y) \right\}$, where $\mathcal{T}_d := \mathcal{T} \setminus \{\tau \in \mathcal{T} \mid \tau(z) \neq d, \forall z \in \mathcal{Z}\}$ and $\tilde{\mathbf{R}}_d$ is $\mathbf{R}_d$ with its zero columns removed. Now, $\tilde{\mathcal{B}}_d(y)$ is a compact convex polytope, and so, $y \mapsto \tilde{\mathcal{B}}_d(y)$ is a polytope bundle, as defined in Billera and Sturmfels (1992). Let $\int \tilde{\mathcal{B}}_d$ be the Minkowski integral of this bundle,

and note that Proposition 1.2 in Billera and Sturmfels (1992) implies that, for any $\mathbf{c} \in \mathbb{R}^{|\mathcal{T}_d|}$, $h_{\int \tilde{\mathcal{B}}_d}(\mathbf{c}) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y)$. Note that if $\tau \in \{\tau \in \mathcal{T} \mid \tau(z) \neq d, \forall z \in \mathcal{Z}\}$, then for any $\mathbf{c} \in \mathbb{R}^{|\mathcal{T}|}$, $\mathbf{c}_{(\tau)} > 0$ implies that

$$h_{\Pi_{\mathcal{T}|d}}(\mathbf{c}) = \infty = \int_{\mathcal{Y}} \infty d\mu(y) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) ,$$

and if $\mathbf{c}_{(\tau)} \leq 0$ for all $\tau \in \{\tau \in \mathcal{T} \mid \tau(z) \neq d, \forall z \in \mathcal{Z}\}$, then letting $\tilde{\mathbf{c}}$ be the projection of $\mathbf{c}$ onto $\mathbb{R}^{|\mathcal{T}_d|}$, we have

$$h_{\Pi_{\mathcal{T}|d}}(\mathbf{c}) = h_{\int \tilde{\mathcal{B}}_d}(\tilde{\mathbf{c}}) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\tilde{\mathbf{c}}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) = \int_{\mathcal{Y}} h_{\tilde{\mathbf{R}}_d}(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) ,$$

which immediately implies the first claim.

For the second claim: Since $\mathbf{R}_d$ does not depend on y , Theorem 1.3 in Billera and Sturmfels (1992) (under the “finite measurable subdivision” definition of piecewise-linear bundle) implies that $\int \tilde{\mathcal{B}}_d$ is a compact convex polytope in $\mathbb{R}_{\geq 0}^{|\mathcal{T}_d|}$, which immediately implies that $\Pi_{\mathcal{T}|d}$ is a convex polytope, since $\Pi_{\mathcal{T}|d}$ is simply the representation of $\int \tilde{\mathcal{B}}_d$ in $\mathbb{R}_{\geq 0}^{|\mathcal{T}|}$. ■

A.4.2. Proof of Lemma 3.2.

Proof. Define $\mathcal{E}_d(y) := \left\{ (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^{|\mathcal{T}|} \times \mathbb{R}_{\geq 0}^{|\mathcal{T}|} \mid \mathbf{x}_1 = \kappa(y) \mathbf{x}_2, \mathbf{R}_d \mathbf{x}_2 = \mathbf{p}\mathbf{f}_d(y) \right\}$

By the same argument as Lemma 3.1, we have that

$$h_{\Theta_{\mathcal{T}|d}(d)}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_{\mathcal{E}_d(y)}(\mathbf{c}_1, \mathbf{c}_2) d\mu(y) .$$

The result now follows by noting that

$$\begin{aligned} h_{\mathcal{E}_d(y)}(\mathbf{c}_1, \mathbf{c}_2) &:= \sup_{(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{E}_d(y)} \mathbf{c}_1^\top \mathbf{x}_1 + \mathbf{c}_2^\top \mathbf{x}_2 \\ &= \sup_{(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{E}_d(y)} \mathbf{c}_1^\top \kappa(y) \mathbf{x}_2 + \mathbf{c}_2^\top \mathbf{x}_2 \\ &= \sup_{\mathbf{x} \in \mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))} (\kappa(y) \mathbf{c}_1 + \mathbf{c}_2)^\top \mathbf{x} \\ &= h_{\mathbf{R}_d}(\kappa(y) \mathbf{c}_1 + \mathbf{c}_2; \mathbf{p}\mathbf{f}_d(y)) . \end{aligned} \quad \blacksquare$$

Appendix B: Proofs of Inference Results

B.1. Proofs of Results related to Specification Test

Proof of Lemma 4.1.

Proof. Here, $\mathbf{R}_d = \mathbf{R}_{d, \mathcal{T}(\mathbb{S})}$, and we have that $\mathcal{B}_{\mathbf{R}_d}(\mathbf{p}\mathbf{f}_d) \neq \emptyset \iff \mathbf{p}\mathbf{f}_d(y) \in \text{cone}(\mathbf{R}_d)$. By the Minkowski-Weyl theorem, every finite cone is a polyhedral cone, i.e. there exists a matrix $\mathbf{A}_d$ such that³³

$$\{\mathbf{b} : \exists \mathbf{x} \geq 0, \mathbf{R}_d \mathbf{x} = \mathbf{b}\} = \{\mathbf{b} : \mathbf{A}_d \mathbf{b} \leq 0\} .$$

³³See, for example, Theorem 3.52 in Rockafellar and Wets (1998). A simple, but highly inefficient, method for finding the matrix $\mathbf{A}_d$ is to use Fourier-Motzkin elimination to eliminate $\mathbf{x}$ from the system $\{\mathbf{R}_d \mathbf{x} = \mathbf{b}, \mathbf{x} \geq 0\}$. More generally, the complexity of finding such a matrix is the same as converting the vertex-representation of a convex polytope to its halfspace representation (indeed, applying such an algorithm to the set of columns of $\mathbf{R}_d$ will yield $\mathbf{A}_d$). For all examples considered in this paper, such a representation can be computed in seconds using `polymake` (free software).

Now, we have that $\mathcal{B}_{\mathbf{R}_d}(\mathbf{p}\mathbf{f}_d) \neq \emptyset \iff \mathbf{A}_d \mathbf{p}\mathbf{f}_d(y) \leq 0$. This restriction represents a set of linear inequalities, so that, for $j \in [|\text{rows}(\mathbf{A}_d)|]$, the linear inequality represented by the j -th row is given by

$$\begin{aligned} & \sum_{z \in \mathcal{Z}} \tilde{c}_{z,j} P(D = d \mid Z = z) f_{P;Y}(y \mid D = d, Z = z) \\ &= \sum_{z \in \mathcal{Z}} \tilde{c}_{z,j} \frac{P(D = d, Z = z)}{P(Z = z)} f_{P;Y}(y \mid D = d, Z = z) \leq 0, \end{aligned}$$

for some coefficients $\{\tilde{c}_{z,j} : z \in \mathcal{Z}\}$.

Define $c_{z,j} := \tilde{c}_{z,j} \left(\prod_{z' \in \mathcal{Z} : z' \neq z} P(Z = z') \right)$. Next, note that

$$\begin{aligned} & \sum_{z \in \mathcal{Z}} \tilde{c}_{z,j} \frac{P(D = d, Z = z)}{P(Z = z)} f_{P;Y}(y \mid D = d, Z = z) \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & \frac{1}{\prod_{z \in \mathcal{Z}} P(Z = z)} \sum_{z \in \mathcal{Z}} c_{z,j} P(D = d, Z = z) f_{P;Y}(y \mid D = d, Z = z) \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & \sum_{z \in \mathcal{Z}} c_{z,j} P(D = d, Z = z) f_{P;Y}(y \mid D = d, Z = z) \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & \sum_{z \in \mathcal{Z}} c_{z,j} P(D = d, Z = z \mid Y = y) f_{P;Y}(y) \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & \sum_{z \in \mathcal{Z}} c_{z,j} P(D = d, Z = z \mid Y = y) \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & \sum_{z \in \mathcal{Z}} c_{z,j} E[\mathbf{1}\{D = d, Z = z\} \mid Y = y] \leq 0, \text{ for all } y \in \mathcal{Y} \\ \iff & E \left[\sum_{z \in \mathcal{Z}} c_{z,j} \mathbf{1}\{D = d, Z = z\} \mid Y = y \right] \leq 0, \text{ for all } y \in \mathcal{Y} \end{aligned}$$

The result now follows by applying the same argument to each row of $\mathbf{A}_d$, and to each $d \in \mathcal{D}$. ■

Proof of Lemma 4.2.

Proof. First assume that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$, for all $y \in \mathcal{Y}$ and $d \in \mathcal{D}$.

Define $\mathcal{T}_d := \{\tau \in \mathcal{T} : \tau(z) = d \text{ for some } z \in \mathcal{Z}\}$. (i.e. all response-types in $\mathcal{T}$ except d -Never-Takers). Assume that $|\mathcal{T}_d| > |\mathcal{Z}|$ (the case $|\mathcal{T}_d| < |\mathcal{Z}|$ is simpler), and that $\mathbf{R}_d$ has full row-rank (to simplify exposition). Assume WLOG that $\mathbf{R}_d$ is such that the d -Never-Takers are the last $|\mathcal{T} \setminus \mathcal{T}_d|$ columns of $\mathbf{R}_d$.

The polytope $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d)$ (which is in $\mathbb{R}^{|\mathcal{T}|}$) can be represented as a polytope in $\mathbb{R}^{|\mathcal{T}_d| - |\mathcal{Z}|}$ defined by all solutions to the system of inequalities

$$\mathbf{M}_d \tilde{\mathbf{x}} \leq \mathbf{b}_d(y),$$

for matrix $\mathbf{M}_d$ and vector $\mathbf{b}_d(y)$ which can be obtained by converting the system of equalities into a system of inequalities in the usual way, via row-reduction and elimination of pivot variables.

Denote by $M_{d,(i)}$ the i -th row of M_d , and define for each row, i , of M_d ,

$$h_{\mathcal{B}_d}(M_{d,(i)}, \mathbf{b}_d(y)) := \sup_{\mathbf{x}: M_d \mathbf{x} \leq \mathbf{b}_d(y)} \{M_{d,(i)}^\top \mathbf{x}\} = h_{\mathbf{R}_d}(M_{d,(i)}; \mathbf{p}\mathbf{f}_d(y)) ,$$

where in the last equality we represent $M_{d,(i)}$ as a vector in $\mathbb{R}^{\mathcal{T}}$ (by setting extra coordinates to 0). Now note that the solutions to

$$M_d \tilde{\mathbf{x}} \leq \begin{bmatrix} h_{\mathcal{B}_d}(M_{d,(1)}, \mathbf{b}_d(y)) \\ \vdots \\ h_{\mathcal{B}_d}(M_{d,(n_d)}, \mathbf{b}_d(y)) \end{bmatrix} ,$$

where n_d is the number of rows in M_d , represents the same polytope as the solutions to $M_d \tilde{\mathbf{x}} \leq \mathbf{b}_d(y)$. The result now follows by repeating this argument for each $d \in \mathcal{D}$, applying Theorem 1.2 in Billera and Sturmfels (1992) along with Proposition 4.1 to obtain the halfspace representation of $\Pi_{\mathcal{T}|d}$, and then applying Theorem 3.3 with the halfspace representation of $\mathcal{F}_{\mathcal{T}}(\mathbb{S})$ given by $S\mathbf{x} \leq \mathbf{s}$, where such a matrix S and vector $\mathbf{s}$ must exist since $\mathbb{S}$ satisfies Condition C.

Finally, the case where $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) = \emptyset$ for some $y \in \mathcal{Y}$ and $d \in \mathcal{D}$ follows automatically for the definitions of M_d , S , and $\mathbf{s}$ given above, since $h_{\mathbf{R}_d}(M_{d,(i)}; \mathbf{p}\mathbf{f}_d(y)) = -\infty$ for all y in some subset with positive measure implies $\int_{\mathcal{Y}} h_{\mathbf{R}_d}(M_{d,(i)}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) = -\infty$. ■

Proof of Proposition 4.1.

Proof. Assume that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$, for all $y \in \mathcal{Y}$ and $d \in \mathcal{D}$ (if not, the result follows immediately from Lemma 4.1), so that $\mathbf{h}_{\mathbb{S}}$ is a real vector. Let the system of inequalities given by Lemma 4.2 be denoted $M\mathbf{x} \leq \mathbf{h}_{\mathbb{S}}$. This system has a solution iff $\mathbf{h}_{\mathbb{S}} \in \text{cone}(M)$, and so, by the Minkowski-Weyl theorem, there exists a matrix A such that the system has a solution iff $A\mathbf{h}_{\mathbb{S}} \leq 0$. As mentioned in the proof of Lemma 4.1, the complexity of finding A is the same as converting a convex polytope from its vertex representation to its halfspace representation. The result now follows immediately from Lemmas 4.1 and 4.2 by defining $\mathbf{c}_i := A_i$. ■

B.2. Proof of Proposition 4.2

Proof. First, note that

$$\Theta_{\mathbb{S}}(\beta) = \{b \in \mathbb{R} \mid \Lambda_{\mathbb{S}}(\beta; b) \cap (\Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \times \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa)) \neq \emptyset\} .$$

Next, define $\Theta := \Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa) \times \Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa)$ and note that the support function of Θ is just the sum of the support functions of $\Theta_{\mathcal{T}(\mathbb{S})|d}(\kappa)$ and $\Theta_{\mathcal{T}(\mathbb{S})|d'}(\kappa)$, evaluated at their respective arguments:

$$\begin{aligned} h_{\Theta}(\mathbf{c}_{(d,1)}, \mathbf{c}_{(d,2)}, \mathbf{c}_{(d',1)}, \mathbf{c}_{(d',2)}) &:= h_{\Theta_{\mathcal{T}|d}(\kappa)}(\mathbf{c}_{(d,1)}, \mathbf{c}_{(d,2)}) + h_{\Theta_{\mathcal{T}|d'}(\kappa)}(\mathbf{c}_{(d',1)}, \mathbf{c}_{(d',2)}) \\ &= \int_{\mathcal{Y}} h_{\mathbf{R}_d}(\kappa(y) \mathbf{c}_{(d,1)} + \mathbf{c}_{(d,2)}; \mathbf{p}\mathbf{f}_d(y)) d\mu(y) \\ &\quad + \int_{\mathcal{Y}} h_{\mathbf{R}_{d'}}(\kappa(y) \mathbf{c}_{(d',1)} + \mathbf{c}_{(d',2)}; \mathbf{p}\mathbf{f}_{d'}(y)) d\mu(y) \end{aligned} \quad (9)$$

Note that Θ is always a non-empty closed convex set when $\mathcal{Q}_S \neq \emptyset$. Now, WLOG³⁴

$$\begin{aligned}
& \gamma \notin \Theta_S(\beta) \\
& \iff \Lambda_S(\beta; \gamma) \cap \Theta = \emptyset \\
& \iff \exists \mathbf{v} \in S^{4|\mathcal{T}|-1}, \sup_{\mathbf{x} \in \Theta} \mathbf{x}^\top \mathbf{v} < \inf_{\mathbf{x} \in \Lambda_S(\beta; \gamma)} \mathbf{x}^\top \mathbf{v} \\
& \iff \exists \mathbf{v} \in S^{4|\mathcal{T}|-1}, h_\Theta(\mathbf{v}) < h_{\Lambda_S(\beta; \gamma)}(-\mathbf{v}) \\
& \iff \exists \mathbf{v} \in S^{4|\mathcal{T}|-1}, 0 < -h_\Theta(\mathbf{v}) - h_{\Lambda_S(\beta; \gamma)}(-\mathbf{v}) \\
& \iff \sup_{\mathbf{v} \in S^{4|\mathcal{T}|-1}} (-h_\Theta(\mathbf{v}) - h_{\Lambda_S(\beta; \gamma)}(-\mathbf{v})) > 0,
\end{aligned}$$

and the result immediately follows by plugging in (9) for h_Θ . ■

Appendix C: Computational Details

C.1. Proof of Proposition 5.1

Proof. (Sorry, this proof is unnecessarily complicated - will fix in future drafts.)

WLOG take $\delta = 0$ since it implies the result for all $\delta > 0$. By $\mathcal{B}_d(\mathbf{pf}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$, we have that $\mathbf{pf}_d(y) \in \text{cone}(\mathbf{R}_d)$ for all y - therefore, the first part of the generic Step j (finding a feasible basis) will terminate. In the remainder, I focus on the second part of the generic Step j (finding an optimal basis).

First, partition $\mathcal{Y}$ into $\mathcal{Y}_1, \dots, \mathcal{Y}_N$ such that each $y, y' \in \mathcal{Y}_i \implies \mathcal{B}_d(\mathbf{pf}_d(y))$ is normally equivalent to $\mathcal{B}_d(\mathbf{pf}_d(y'))$. Since $\mathbf{pf}_d$ is continuous (and $\mathcal{Y}$ is compact), we can take the partition to be defined by numbers

$$\min \mathcal{Y} =: y_0 \leq \dots \leq y_N := \max \mathcal{Y},$$

such that³⁵

$$\mathcal{Y} = [y_0, y_1] \sqcup \left(\bigsqcup_{i=1}^{N-1} (y_i, y_{i+1}] \right).$$

By definition, for y within each partition $\mathcal{Y}_k$, $\mathcal{B}_d(\mathbf{pf}_d(y))$ has the same normal fan, denote it $\mathcal{N}_k$, which is a decomposition of $\mathbb{R}^{|\mathcal{T}|}$ into polyhedral cones, $\mathcal{N}_{k,1}, \dots, \mathcal{N}_{k,N_k}$.

Now, further partition each cell $\mathcal{Y}_k$ of the above partition into $\mathcal{Y}_{k,1}, \dots, \mathcal{Y}_{k,N_k}$ such that $y, y' \in \mathcal{Y}_{k,j}$ implies that $\varkappa(y), \varkappa(y') \in \mathcal{N}_{k,j}$. Since $\varkappa$ is a continuous function over a compact set, we can once again WLOG take these partitions to be intervals³⁶.

Now, we have partitioned $\mathcal{Y}$ into finitely many intervals such that the LP

$$\max \varkappa(y)^\top \mathbf{x} \quad \text{subject to} \quad \mathbf{R}_d \mathbf{x} = \mathbf{pf}_d(y), \mathbf{x} \geq \mathbf{0},$$

will have the same optimal base(s) for all y within each cell of the partition. This shows the final claim.

³⁴If γ is such that $\Lambda_S(\beta; \gamma)$ is non-empty, the test automatically rejects the null (as desired) since the last term will resolve to $-(-\infty) = \infty$, making the whole expression resolve to ∞ for any $\mathbf{v}$.

³⁵Technical point: to represent the partitions in this way, we can relax the requirement of normal equivalence for points on the boundary, so that they are only required to coarsen the normal fan associated with the interior points of the partition. The intuition is the same, however.

³⁶This may lead to potentially more than N_k partitions for each $\mathcal{Y}_k$, but simply redefine N_k to accommodate this, if necessary.

Now, it suffices to show that there will always exist some $n < \infty$ such that, for all cells (k, j) , either $\mathcal{G}_n \cap \text{int}(\mathcal{Y}_{k,j}) \neq \emptyset$ or $\mathcal{G}_n \cap \mathcal{Y}_{k,j} \neq \emptyset$ and the LP has the same optimal base for $y \in \mathcal{G}_n \cap \mathcal{Y}_{k,j}$ as it does for points on the interior.

This follows immediately by noting that if n is such that this condition does not hold for some cell(s), then points in all such cells will have positive duality gap, and so, $\mathcal{G}_{n+1}$ will necessarily satisfy the condition for at least one more cell. Since there are only finitely many cells, this argument implies termination in finitely many steps. This shows the first claim in (i). (ii) and the second claim in (i) are immediate consequences of the weak duality theorem, completing the proof. ■

Appendix D: Proofs of Auxiliary Results

Proposition 4.1. *Let P_1 and P_2 be bounded full-dimensional polytopes in $\mathbb{R}^n$ defined as $\{x \geq 0 : Ax \leq b_1\}$ and $\{x \geq 0 : Ax \leq b_2\}$, respectively. Denote by h_{P_i} the support function of P_i . Let $\{a_1, \dots, a_N\}$ be the rows of A , let $\bar{b}_1 := (h_{P_1}(a_i) : i \in \{1, \dots, N\})$, and let $\bar{b}_2 := (h_{P_2}(a_i) : i \in \{1, \dots, N\})$. Then,*

$$P_1 + P_2 = \{x \geq 0 : Ax \leq \bar{b}_1 + \bar{b}_2\}.$$

Proof. Denote $\bar{P} := \{x \geq 0 : Ax \leq \bar{b}_1 + \bar{b}_2\}$.

First, note that $h_{P_1+P_2}(a) = h_{P_1}(a) + h_{P_2}(a)$ for all $a \in \mathbb{R}^n$, and so, $P_1 + P_2 \subseteq \bar{P}$.

Next, let ∂P denote the boundary of P , for any polytope P . We have that

$$\partial(P_1 + P_2) = \bigcup_{i=1}^N \{x_1 + x_2 : x_1 \in P_1, x_2 \in P_2; a'_i x_1 = h_{P_1}(a_i), a'_i x_2 = h_{P_2}(a_i)\}.$$

Now, note that if $(x_1 + x_2) \in \partial(P_1 + P_2)$ such that $a'_i x_1 = h_{P_1}(a_i)$, $a'_i x_2 = h_{P_2}(a_i)$, then necessarily $a'_i(x_1 + x_2) = h_{P_1}(a_i) + h_{P_2}(a_i) = \bar{b}_1 + \bar{b}_2$, and so, $x_1 + x_2 \in \partial \bar{P}$. Therefore, we have that $\partial(P_1 + P_2) \subseteq \partial \bar{P}$.

Next, note that both $P_1 + P_2$ and $\bar{P}$ have the same dimension, n , and so, their boundaries are both homeomorphic to S^{n-1} , the unit sphere in $\mathbb{R}^{n-1}$ ³⁷. Let f be the homeomorphism between $\bar{P}$ and S^{n-1} .

Suppose that $\partial(P_1 + P_2) \neq \partial \bar{P}$, then $f|_{\partial(P_1+P_2)}$ is a homeomorphism between $\partial(P_1 + P_2)$ and $f(\partial(P_1 + P_2)) \subsetneq S^{n-1}$. Since $\partial(P_1 + P_2)$ is itself homeomorphic to S^{n-1} , this implies that S^{n-1} is homeomorphic to its strict subset $f(\partial(P_1 + P_2))$. This is a contradiction, however, since the only subset of S^{n-1} that is homeomorphic to S^{n-1} is S^{n-1} itself. Therefore, $\partial(P_1 + P_2) = \partial \bar{P}$, which immediately implies that $\bar{P} = P_1 + P_2$, as required³⁸. ■

Corollary 4.1. *Let P_1 and P_2 be bounded full-dimensional polytopes in $\mathbb{R}^n$ defined as $\{x \geq 0 : Ax \leq b_1\}$ and $\{x \geq 0 : Bx \leq b_2\}$, respectively. Let $A = \begin{pmatrix} A' \\ C \end{pmatrix}$ and let $B = \begin{pmatrix} B' \\ C \end{pmatrix}$, where C is the matrix consisting of (potentially zero) rows that are shared between A and B . Let $\{d_1, \dots, d_N\}$ be the rows of $\begin{pmatrix} A' \\ B' \\ C \end{pmatrix} =: D$, and let $\bar{b}_1 := (h_{P_1}(d_i) : i \in \{1, \dots, N\})$, and let $\bar{b}_2 := (h_{P_2}(d_i) : i \in \{1, \dots, N\})$. Then,*

$$P_1 + P_2 = \{x \geq 0 : Dx \leq \bar{b}_1 + \bar{b}_2\}.$$

³⁷If $K \subseteq \mathbb{R}^n$ is convex, compact, and has non-empty interior, then its boundary is homeomorphic to S^{n-1} .

³⁸Special thanks to Eric Wofsey for the idea used to prove $\partial(P_1 + P_2) \subseteq \partial \bar{P} \implies \partial(P_1 + P_2) = \partial \bar{P}$.

Proof. Follows immediately from Proposition 4.1 by noting that

$$P_1 = \{x \geq 0 : Dx \leq \bar{b}_1\} \quad \text{and} \quad P_2 = \{x \geq 0 : Dx \leq \bar{b}_2\} . \blacksquare$$

Appendix E: Equivalence of Structural IV Model and Potential Outcomes Model

I assume that the probability space (Ω, Σ, P) is sufficiently enriched for the applications of the Skorohod's theorem used in this section. This assumption can always be satisfied after suitably enriching (Ω, Σ, P) , if needed, by taking product spaces.³⁹

Condition PO (Potential Outcomes Model). *There exists random variables $((Y_{d,z} : d \in \mathcal{D}, z \in \mathcal{Z}), (D_z : z \in \mathcal{Z}))$ defined on (Ω, Σ, P) such that:*

(i) *For all $z, z' \in \mathcal{Z}$ and $d \in \mathcal{D}$, $Y_d := Y_{d,z} = Y_{d,z'}$ almost surely. Further,*

$$D = \sum_{z \in \mathcal{Z}} \mathbb{1}\{Z = z\} D_z ,$$

$$Y = \sum_{d \in \mathcal{D}} \mathbb{1}\{D = d\} Y_d ,$$

almost surely.

(ii) $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z$

Proposition 5.1. *Let P be a distribution of the observed data, (Y, D, Z) . P satisfies Condition PO if and only if it satisfies Condition M.*

Proof. First, suppose P satisfies Condition M. Define

$$Y_d := g_Y(d, \mathbf{V}, \varepsilon_Y) ,$$

$$D_z := g_D(z, \mathbf{V}) ,$$

which implies that P satisfies Condition PO(i). Next, note that Condition M(ii) implies that

$$Z, \mathbf{V}, \varepsilon_Y \text{ are mutually independent}$$

$$\implies ((g_Y(d, \mathbf{V}, \varepsilon_Y) : d \in \mathcal{D}), (g_D(z, \mathbf{V}) : z \in \mathcal{Z})) \perp Z$$

$$\implies ((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \perp Z ,$$

which implies that P satisfies Condition PO.

Now, suppose P satisfies Condition PO, and let $((Y_d : d \in \mathcal{D}), (D_z : z \in \mathcal{Z})) \sim Q$ be random variables that witness this; note that $P_Q = P$. Define $\mathbf{V} = (D_z : z \in \mathcal{Z})$. For each $\mathbf{v} \in \text{supp}_Q(\mathbf{V})$, define $\varepsilon_Y(\mathbf{v})$ as a $|\mathcal{D}|$ -dimensional random vector with distribution $Q_{(Y_d : d \in \mathcal{D}) | \mathbf{V} = \mathbf{v}}$, with $\varepsilon_Y := (\varepsilon_Y(\mathbf{v})^\top : \mathbf{v} \in \text{supp}_Q(\mathbf{V}))^\top$ independent from $(Z, \mathbf{V})$. By an application of the Skorohod representation theorem,⁴⁰ we can take ε_Y to be such that, for every $\mathbf{v} \in \text{supp}(\mathbf{V})$, $\mathbb{1}\{\mathbf{V} = \mathbf{v}\} \varepsilon_Y(\mathbf{v}) = (\mathbb{1}\{\mathbf{V} = \mathbf{v}\} Y_d : d \in \mathcal{D})$ almost surely.

Next, define $g_D : \mathcal{Z} \times \text{supp}_Q(\mathbf{V})$ and $g_Y : \mathcal{D} \times \text{supp}(\mathbf{V}) \times \text{supp}(\varepsilon_Y)$ as follows

$$g_D(z, \mathbf{v}) := \mathbf{v}_{(z)} ,$$

$$g_Y(d, \mathbf{v}, \mathbf{e}) := \mathbf{e}_{(\mathbf{v})(d)} ,$$

where $\mathbf{e}_{(\mathbf{v})(d)}$ is the d -th component of the $\mathbf{v}$ -th tuple of the stacked tuple $\mathbf{e} \in \text{supp}(\varepsilon_Y)$. Now, the result follows immediately by noting that $(g_Y(g_D(Z, \mathbf{V}), \mathbf{V}, \varepsilon_Y), g_D(Z, \mathbf{V}), Z) = (Y, D, Z)$ almost surely. $\blacksquare$

³⁹See Halmos (1976), p. 157.

⁴⁰See, e.g., Billingsley (1999), p. 70.